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MODERN PHYSICS_MEGA_XII



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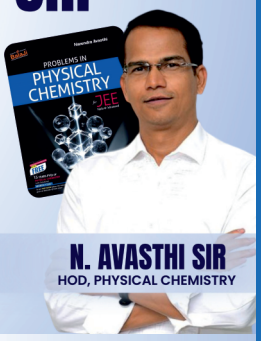

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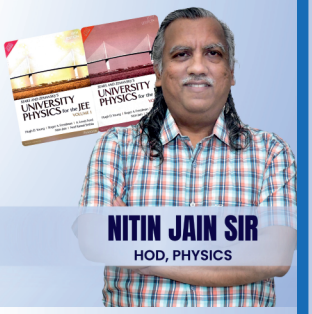


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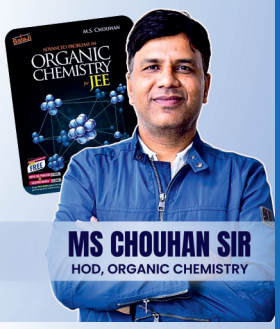


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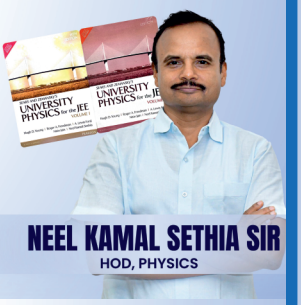


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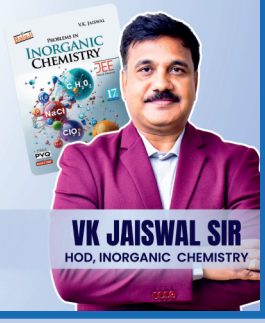


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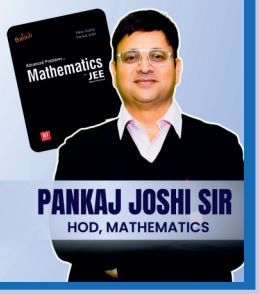


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PHYSICS

TARGET IIT JEE

MEGA-XII

MODERN PHYSICS

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KEY CONCEPT

1. CATHODE RAYS :

(a) Generated in a discharge tube in which a high vaccum is maintained.

(b) They are electrons accelerated by high p.d. (10 to 15 K.V.)

(c) K.E. of C.R. particle accelerated by a p.d. V is $\frac{1}{2}mv^2 = \frac{P^2}{2m}$ = eV.

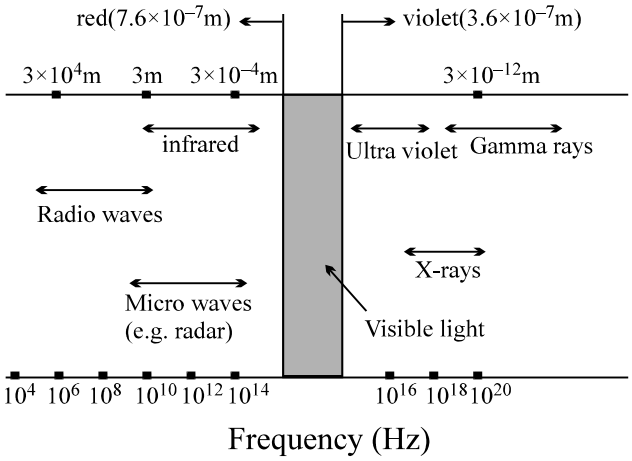
(d) Can be deflected by Electric & magnetic fields .

2. ELECTROMAGNETIC SPECTRUM :

Ordered arrangement of the big family of electro magnetic waves (EMW) either in ascending order of frequencies or of wave lengths

Speed of E.M.W. in vacuum

$C = 3 \times 10^8$ m/s = $\nu\lambda$



3. PLANK'S QUANTUM THEORY :

A beam of EMW is a stream of discrete packets of energy called **PHOTONS**, each photon having a frequency ν and energy = E = h ν.

h = plank's constant = 6.63×10^{-34} Js.

4. PHOTO ELECTRIC EFFECT :

The phenomenon of the emission of electrons, when metals are exposed to light (of a certain minimum frequency) is called photo electric effect.

Results :

(i) Can be explained only on the basis of the quantum theory (concept of photon).

(ii) Electrons are emitted if the incident light has frequency $\nu \geq \nu_0$ (threshold frequency) emission of electrons is independent of intensity. The wave length corresponding to ν_0 is called threshold wave length λ_0.

(iii) ν_0 is different for different metals .

(iv) Number of electrons emitted per second depends on the intensity of the incident light.

(v) EINSTEINS PHOTO ELECTRIC EQUATION :

Photon energy = K. E. of electron + work function.

$$h\nu = \frac{1}{2}mv^2 + \phi$$

ϕ = **Work function = energy needed by the electron in freeing itself from the atoms of the metal .**

$$\phi = h\,\nu_0$$

(vi) STOPPING POTENTIAL OR CUT OFF POTENTIAL :

The minimum value of the retarding potential to prevent electron emission is :

$$eV_{cut\ off} = (KE)_{max}$$

Note : The number of photons incident on a surface per unit time is called photon flux.

5. WAVE NATURE OF MATTER :

Beams of electrons and other forms of matter exhibit wave properties including interference and diffraction with a de Broglie wave length given by $\lambda = \frac{h}{p}$

(wave length of a praticle) .

6. ATOMIC MODELS :

(a) THOMSON MODEL : (PLUM PUDDING MODEL)

(i) Most of the mass and all the positive charge of an atom is uniformly distributed over the full size of atom (10^{-10} m).

(ii) Electrons are studded in this uniform distribution .

(iii) Failed to explain the large angle scattering α-particle scattered by thin foils of matter .

(b) RUTHERFORD MODEL : (Nuclear Model)

(i) The most of the mass and all the positive charge is concentrated within a size of 10^{-14} m inside the atom. This concentration is called the atomic nucleus.

(ii) The electron revolves around the nucleus under electric interaction between them in circular orbits. An accelerating charge radiates the nucleus spiralling inward and finally fall into the nucleus, which does not happen in an atom. This could not be explained by this model.

(c) BOHR ATOMIC MODEL :

Bohr adopted Rutherford model of the atom & added some arbitrary conditions. These conditions are known as his postulates :

(i) The electron in a stable orbit does not radiate energy . i.e. $\frac{mv^2}{r}=\frac{kze^2}{r^2}$

(ii) A stable orbit is that in which the angular momentum of the electron about nucleus is an integral (n) multiple of $\frac{h}{2\pi}$. i.e. $mvr = n\frac{h}{2\pi}$; n = 1 , 2 , 3 , ...($n \neq 0$).

(iii) The electron can absorb or radiate energy only if the electron jumps from a lower to a higher orbit or falls from a higher to a lower orbit .

(iv) The energy emitted or absorbed is a light photon of frequency ν and of energy . **$E = h\nu$** .

FOR HYDROGEN ATOM : (Z = atomic number = 1)

(i) L_n = angular momentum in the nth orbit = $n\frac{h}{2\pi}$.

(ii) r_n = radius of nth circular orbit = $(0.529\ A^o)\ n^2$; $(1A^o = 10^{-10}\ m)$; $r_n \alpha\ n^2$.

(iii) E_n Energy of the electron in the nth orbit = $\frac{-13.6\ ev}{n^2}$ i.e. $E_n\ \alpha\ \frac{1}{n^2}$.

Note : Total energy of the electron in an atom is negative, indicating that it is bound.

Binding Energy $(BE)_n = -E_n = \frac{13.6\ ev}{n^2}$.

(iv) $E_{n2} - E_{n1}$ = Energy emitted when an electron jumps from n_2^{th} orbit to n_1^{th} orbit $(n_2 > n_1)$.

$\Delta E = (13.6\ ev)\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$.

$\Delta E = h\nu$; ν = frequency of spectral line emitted .

$\frac{1}{\lambda} = \nu$ = wave no. [no. of waves in unit length (1m)] = $R\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$.

Where R = Rydberg's constant for hydrogen = $1.097 \times 10^7\ m^{-1}$.

(v) For hydrogen like atom/spicies of atomic number Z :

$r_{nz} = \frac{\text{Bohr radius}}{Z}\ n^2 = (0.529\ A^o)\ \frac{n^2}{Z}$; $E_{nz} = (-13.6)\ \frac{Z^2}{n^2}\ ev$

$R_z = RZ^2$ – Rydberg's constant for element of atomic no. Z .

Note : If motion of the nucleus is also considered, then m is replaced by μ .
Where μ = reduced mass of electron - nucleus system = $mM/(m + M)$.

In this case $E_n = (-13.6\ ev)\ \frac{Z^2}{n^2}.\frac{\mu}{m_e}$

7. SPECTRAL SERIES :

(i) Lyman Series : (Landing orbit n = 1) .

Ultraviolet region $\bar{\nu} = R\left[\frac{1}{1^2} - \frac{1}{n_2^2}\right]$; $n_2 > 1$

(ii) Balmer Series : (Landing orbit n = 2)

Visible region $\bar{\nu} = R\left[\frac{1}{2^2} - \frac{1}{n_2^2}\right]$; $n_2 > 2$

(iii) Paschan Series : (Landing orbit n = 3)

In the near infrared region $\bar{\nu} = R\left[\frac{1}{3^2} - \frac{1}{n_2^2}\right]$; $n_2 > 3$

(iv) Bracket Series : (Landing orbit n = 4)

In the mid infrared region $\bar{\nu} = R\left[\frac{1}{4^2} - \frac{1}{n_2^2}\right]$; $n_2 > 4$

(v) Pfund Series : (Landing orbit n = 5)

In far infrared region $\bar{\nu} = R\left[\frac{1}{5^2} - \frac{1}{n_2^2}\right]$; $n_2 > 5$

In all these series n_2 = $n_1 + 1$ is the α line
= $n_1 + 2$ is the β line
= $n_1 + 3$ is the γ line ..... etc . where n_1 = Landing orbit

8. EXCITATION POTENTIAL OF ATOM :

Excitation potential for quantum jump from $n_1 \longrightarrow n_2 = \frac{E_{n2} - E_{n1}}{\text{electron charge}}$.

9. IONIZATION ENERGY :

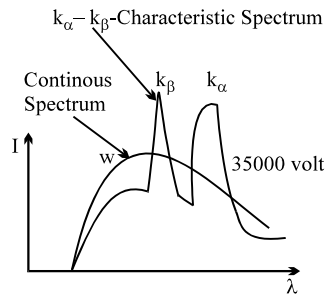
The energy required to remove an electron from an atom. The energy required to ionize hydrogen atom is = 0 – (–13.6) = 13.6 ev .

10. IONIZATION POTENTIAL :

Potential difference through which an electron is moved to gain ionization energy = $\frac{-E_n}{\text{electronic charge}}$

11. X - RAYS :

(i) Short wavelength (0.1 A° to 1 A°) electromagnetic radiation .
(ii) Are produced when a metal anode is bombarded by very high energy electrons .
(iii) Are not affected by electric and magnetic field .
(iv) They cause photoelectric emission .
Characteristics equation $eV = h\nu_m$
e = electron charge ;
V = accelerating potential
ν_m = maximum frequency of X - radiation



(v) Intensity of X - rays depends on number of electrons hitting the target .
(vi) Cut off wavelength or minimum wavelength,

where v (in volts) is the p.d. applied to the tube $\lambda_{min} \cong \frac{12400}{V}$ A° .

(vii) Continuous spectrum due to retardation of electrons.
(viii) Characteristic Spectrum due to transition of electron from higher to lower
$\nu \propto (z - b)^2$; $\upsilon = a(z - b)^2$ **[MOSELEY'S LAW]**
b = 1 for k_α transition

Note : **(i)** Binding energy = –[Total Mechanical Energy]

(ii) Vel. of electron in n^{th} orbit for hydrogen atom $\cong \frac{c}{137n}$; c = speed of light .

(iii) For x-rays $\frac{1}{\lambda} = R(z-b)^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

(iv) Series limit of series means minimum wave length of that series.

12. NUCLEAR DIMENSIONS :

$R = R_o A^{1/3}$ Where R_o = empirical constant = 1.1×10^{-15} m ; A = Mass number of the atom

13. RADIOACTIVITY :

The phenomenon of self emission of radiation is called radioactivity and the substances which emit these radiations are called **radioactive substances** . It can be natural or artificial (induced) .

14. α , β , γ RADIATION :

(i) α-particle :

(a) Helium nucleus ($_2He^4$) ; **(b)** energy varies from 4 Mev to 9 Mev ;

(c) Velocity $10^6 - 10^7$ m/s ; **(d)** low penetration

(ii) β-particle :

(a) Have much less energy ; **(b)** more penetration ; **(c)** higher velocities than α particles

(iii) γ-radiation : Electromagnetic waves of very high energy .

15. LAWS OF RADIOACTIVE DISINTEGRATION :

(A) **DISPLACEMENT LAW** : In all radioactive transformation either an α or β particle (never both or more than one of each simultaneously) is emitted by the nucleus of the atom.

(i) α – emission : $_zX^A \longrightarrow {}_{z-2}Y^{A-4} + {}_2\alpha^4 +$ Energy

(ii) β – emission : $_zX^A \longrightarrow \beta + {}_{z+1}Y^A + \overline{\nu}$ (antinuetrino)

(iii) γ – emission : emission does not affect either the charge number or the mass number.

(B) **STASTISTICAL LAW** : The disintegration is a random phenomenon. Whcih atom disintegrates first is purely a matter of chance.

Number of nuclei disintegrating per second is given ;

(disintegration /s /gm is called specific activity).

(i) $\frac{dN}{dt} \alpha N \rightarrow \frac{dN}{dt} = -\lambda N$ = activity .

Where N = No. of nuclei present at time t ; λ = decay constant

(ii) $N = N_o e^{-\lambda t}$ N_o = number of nuclei present in the beginning .

(iii) Half life of the population $T_{1/2} = \frac{0.693}{\lambda}$;

at the end of n half–life periods the number of nuclei left $N = \frac{N_o}{2^n}$.

(iv) MEAN LIFE OF AN ATOM = $\frac{\Sigma \text{ life time of all atoms}}{\text{total number of atoms}}$; $T_{av} = \frac{1}{\lambda}$

(v) **CURIE :** The unit of activity of any radioactive substance in which the number of disintegration per second is 3.7×10^{10}.

16. ATOMIC MASS UNIT (a.m.u. OR U) :

$1\text{ amu} = \frac{1}{12} \times (\text{mass of carbon} - 12\text{ atom}) = 1.6603 \times 10^{-27}\text{ kg}$

17. MASS AND ENERGY :

The mass m of a particle is equivalent to an energy given by $E = mc^2$;
c = speed of light . 1 amu = 931 Mev

18. MASS DEFECT AND BINDING ENERGY OF A NUCLEUS :

The nucleus is less massive than its constituents. The difference of masses is called mass defect.
ΔM = mass defect = $[Z_{mp} + (A - Z)\, m_n] - M_{zA}$.

Total energy required to be given to the nucleus to tear apart the individual nucleons composing the nucleus , away from each other and beyond the range of interaction forces is called the Binding Energy of a nucleus .
B.E. = $(\Delta M)C^2$.

B.E. per nucleon = $\frac{(\Delta M)C^2}{A}$.

Greater the B.E., greater is the stability of the nucleus .

19. NUCLEAR FISSION :

(i) Heavy nuclei of A , above 200 , break up onto two or more fragments of comparable masses.
(ii) The total B.E. increases and excess energy is released .
(iii) The main point of the fission energy is leberated in the form of the K.E. of the fission fragments.

eg. ${}^{235}_{92}U + {}_0n^1 \rightarrow {}^{236}_{92}U \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3\ {}_0n^1$ + energy

20. NUCLEAR FUSION (Thermo nuclear reaction) :

(i) Light nuclei of A below 20, fuse together, the B.E. per nucleon increases and hence the excess energy is released.
(ii) These reactions take place at ultra high temperature ($\cong 10^7$ to 10^9)
(iii) Energy released exceeds the energy liberated in the fission of heavy nuclei.

eg . ${}^{1}_{1}P \rightarrow {}^{4}_{1}He + {}^{0}_{+1}e$. (Positron)

(iv) The energy released in fusion is specified by specifying Q value.
i.e. Q value of reaction = energy released in a reaction.

Note : **(i)** In emission of β^- , z increases by 1.
(ii) In emission of β^+ , z decreases by 1.

EXERCISE-I

ONLY ONE OPTION IS CORRECT

1. If h is Planck's constant is SI system, the momentum of a photon of wavelength 0.01 Å is:
(A) 10^{-2} h (B) h (C) 10^{2} h (D) 10^{12} h

2. When a photon of light collides with a metal surface, number of electrons, (if any) coming out is
(A) only one (B) only two
(C) infinite (D) depends upon factors

3. 10^{-3} W of 5000 Å light is directed on a photoelectric cell. If the current in the cell is 0.16 μA, the percentage of incident photons which produce photoelectrons, is
(A) 0.4% (B) 0.04% (C) 20% (D) 10%

4. Let n_r and n_b be respectively the number of photons emitted by a red bulb and a blue bulb of equal power in a given time.
(A) $n_r = n_b$ (B) $n_r < n_b$
(C) $n_r > n_b$ (D) data insufficient

5. In a photo-emissive cell, with exciting wavelength λ, the maximum kinetic energy of electron is K. If the exciting wavelength is changed to $\frac{3\lambda}{4}$ the kinetic energy of the fastest emitted electron will be:
(A) 3K/4 (B) 4K/3
(C) less than 4K/3 (D) greater than 4K/3

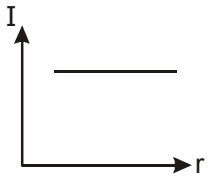
6. If the frequency of light in a photoelectric experiment is doubled, the stopping potential will
(A) be doubled (B) halved
(C) become more than doubled (D) become less than double

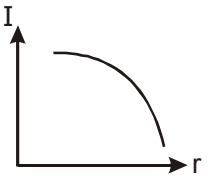
7. The stopping potential for the photo electrons emitted from a metal surface of work function 1.7 eV is 10.4 V. Identify the energy levels corresponding to the transitions in hydrogen atom which will result in emission of wavelength equal to that of incident radiation for the above photoelectric effect
(A) n = 3 to 1 (B) n = 3 to 2 (C) n = 2 to 1 (D) n = 4 to 1

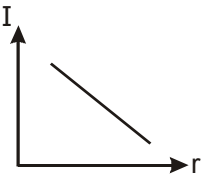
8. The frequency and the intensity of a beam of light falling on the surface of photoelectric material are increased by a factor of two. Treating efficiency of photoelectron generation as constant, this will :
(A) increase the maximum energy of the photoelectrons, as well as photoelectric current by a factor of two.
(B) increase the maximum kinetic energy of the photo electrons and would increase the photoelectric current by a factor of two.
(C) increase the maximum kinetic energy of the photoelectrons by a factor of greater than two and will have no effect on the magnitude of photoelectric current produced.
(D) not produce any effect on the kinetic energy of the emitted electrons but will increase the photoelectric current by a factor of two.

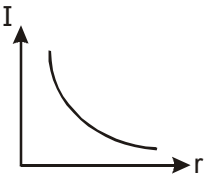
9. A point source of ligth is used in a photoelectric effect. If the source is removed farther from the emitting metal, the stopping potential :
(A) will increase (B) will decrease
(C) will remain constant (D) will either increase or decrease.

10. A point source causes photoelectric effect from a small metal plate. Which of the following curves may represent the saturation photocurrent as a function of the distance between the source and the metal?

(A) 

(B) 

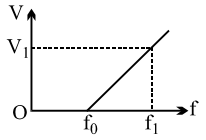
(C) 

(D) 

11. Let K_1 be the maximum kinetic energy of photoelectrons emitted by a light of wavelength λ_1 and K_2 corresponding to λ_2. If $\lambda_1 = 2\lambda_2$, then :

(A) $2K_1 = K_2$ (B) $K_1 = 2K_2$ (C) $K_1 < \frac{K_2}{2}$ (D) $K_1 > 2K_2$

12. In a photoelectric experiment, the potential difference V that must be maintained between the illuminated surface and the collector so as just to prevent any electron from reaching the collector is determined for different frequencies f of the incident illumination. The graph obtained is shown. The maximum kinetic energy of the electrons emitted at frequency f_1 is



(A) hf_1 (B) $\frac{V_1}{(f_1 - f_0)}$

(C) $h(f_1 - f_0)$ (D) $eV_1(f_1 - f_0)$

13. Radiation of two photon energies twice and five times the work function of metal are incident sucessively on the metal surface. The ratio of the maximum velocity of photoelectrons emitted is the two cases will be

(A) 1 :2 (B) 2 : 1 (C) 1 : 4 (D) 4 : 1

14. Cut off potentials for a metal in photoelectric effect for light of wavelength λ_1, λ_2 and λ_3 is found to be V_1, V_2 and V_3 volts if V_1, V_2 and V_3 are in Arithmetic Progression and λ_1, λ_2 and λ_3 will be:

(A) Arithmetic Progression (B) Geometric Progression

(C) Harmonic Progression (D) None

15. Photons with energy 5 eV are incident on a cathode C, on a photoelectric cell. The maximum energy of the emitted photoelectrons is 2 eV. When photons of energy 6 eV are incident on C, no photoelectrons will reach the anode A if the stopping potential of A relative to C is

(A) 3 V (B) –3V (C) –1V (D) 4 V

16. In a photoelectric experiment, the collector plate is at 2.0V with respect to the emitter plate made of copper (φ = 4.5 eV). The emitter is illuminated by a source of monochromatic light of wavelength 200 nm.

(A) the minimum kinetic energy of the photoelectrons reaching the collector is 0.

(B) the maximum kinetic energy of the photoelectrons reaching the collector is 3.7 eV.

(C) if the polarity of the battery is reversed then answer to part A will be 0.

(D) if the polarity of the battery is reversed then answer to part B will be 1.7 eV.

17. By increasing the intensity of incident light keeping frequency ($v > v_0$) fixed on the surface of metal

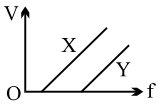
(A) kinetic energy of the photoelectrons increases

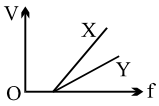
(B) number of emitted electrons increases

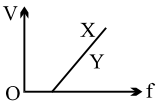
(C) kinetic energy and number of electrons increases

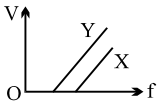
(D) no effect

18. In a photoelectric experiment, electrons are ejected from metals X and Y by light of intensity I and frequency f. The potential difference V required to stop the electrons is measured for various frequencies. If Y has a greater work function than X ; which one of the following graphs best illustrates the expected results?

(A) 

(B) 

(C) 

(D) 

19. Monochromatic light with a frequency well above the cutoff frequency is incident on the emitter in a photoelectric effect apparatus. The frequency of the light is then doubled while the intensity is kept constant. How does this affect the photoelectric current?

(A) The photoelectric current will increase. (B) The photoelectric current will decrease.

(C) The photoelectric current will remain the same. (D) None of these

20. Imagine a Young's double slit interference experiment performed with waves associated with fast moving electrons produced from an electron gun. The distance between successive maxima will decrease maximum if

(A) the accelerating voltage in the electron gun is decreased

(B) the accelerating voltage is increased and the distance of the screen from the slits is decreased

(C) the distance of the screen from the slits is increased.

(D) the distance between the slits is decreased.

21. A proton and an electron are accelerated by same potential difference have de-Broglie wavelength λ_p and λ_e.

(A) $\lambda_e = \lambda_p$ (B) $\lambda_e < \lambda_p$ (C) $\lambda_e > \lambda_p$ (D) none of these.

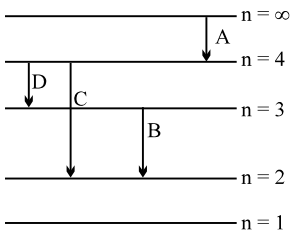
22. Consider the following electronic energy level diagram of H-atom:
Photons associated with shortest and longest wavelengths would be emitted from the atom by the transitions labelled:

(A) D and C respectively

(B) C and A respectively

(C) C and D respectively

(D) A and C respectively



23. The angular momentum of an electron in the hydrogen atom is $\frac{3h}{2\pi}$. Here h is Planck's constant. The kinetic energy of this electron is:

(A) 4.53 eV (B) 1.51 eV (C) 3.4 eV (D) 6.8 eV

24. The binding energies of nuclei X and Y are E_1 and E_2 respectively. Two atoms of X fuse to give one atom of Y and an energy Q is released. Then :

(A) $Q = 2E_1 - E_2$ (B) $Q = E_2 - 2E_1$ (C) $Q = 2E_1 + E_2$ (D) $Q = 2E_2 + E_1$

25. If the electron in a hydrogen atom were in the energy level with n = 3, how much energy in joule would be required to ionise the atom? (Ionisation energy of H-atom is 2.18×10^{-18}J) :

(A) 6.54×10^{-19} (B) 1.43×10^{-19} (C) 2.42×10^{-19} (D) 3.14×10^{-20}

26. If radiation of all wavelengths from ultraviolet to infrared is passed through hydrogen gas at room temperature, absorption lines will be observed in the :

(A) Lyman series (B) Balmer series (C) both (A) and (B) (D) neither (A) nor (B)

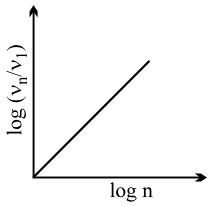
27. In hydrogen and hydrogen like atoms, the ratio of difference of energies $E_{4n} - E_{2n}$ and $E_{2n} - E_n$ varies with its atomic number z and n as :

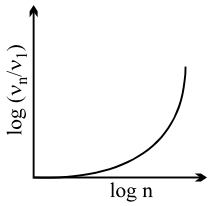
(A) z^2 / n^2 (B) z^4/n^4 (C) z / n (D) $z^0 n^0$

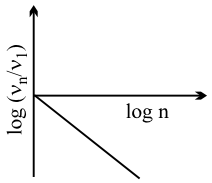
28. In a hydrogen atom, the electron is in nth excited state. It may come down to second excited state by emitting ten different wavelengths. What is the value of n :

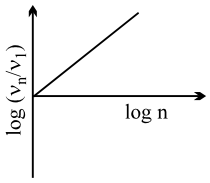
(A) 6 (B) 7 (C) 8 (D) 5

29. The frequency of revolution of electron in n^{th} Bohr orbit is ν_n. The graph between log n and log (ν_n / ν_1) may be

(A) 

(B) 

(C) 

(D) 

30. Difference between nth and $(n + 1)^{th}$ Bohr's radius of 'H' atom is equal to it's (n – 1)th Bohr's radius. the value of n is:

(A) 1 (B) 2 (C) 3 (D) 4

31. In an atom, two electrons move around the nucleus in circular orbits of radii R and 4R. The ratio of the time taken by them to complete one revolution is : (neglect electric interaction)

(A) 1 : 4 (B) 4 : 1 (C) 1 : 8 (D) 8 : 1

32. The electron in a hydrogen atom makes transition from M shell to L. The ratio of magnitudes of initial to final centripetal acceleration of the electron is

(A) 9 : 4 (B) 81 : 16 (C) 4 : 9 (D) 16 : 81

33. The radius of Bohr's first orbit is a_0. The electron in n^{th} orbit has a radius:

(A) na_0 (B) a_0/n (C) n^2a_0 (D) a_0/n^2

34. The magnitude of angular momentum, orbit radius and frequency of revolution of electron in hydrogen atom corresponding to quantum number n are L, r and f respectively. Then according to Bohr's theory of hydrogen atom,

(A) fr^2L is constant for all orbits (B) frL is constant for all orbits

(C) f^2rL is constant for all orbits (D) frL^2 is constant for all orbits

35. The electron in a hydrogen atom make a transition from an excited state to the ground state. Which of the following statement is true?

(A) Its kinetic energy increases and its potential and total energies decrease

(B) Its kinetic energy decreases, potential energy increases and its total energy remains the same.

(C) Its kinetic and toal energies decrease and its potential energy increases.

(D) its kinetic potential and total energies decreases.

36. The ionisation potential of hydrogen atom is 13.6 volt. The energy required to remove an electron from the second orbit of hydrogen is :

(A) 3.4 eV (B) 6.8 eV (C) 13.6 eV (D) 27.2 eV

37. In a hypothetical system a particle of mass m and charge –3q is moving around a very heavy particle having cahrge q. Assuming Bohr's model to be true to this system, the orbital velocity of mass m when it is nearest to heavy particle is

(A) $\frac{3q^2}{2\varepsilon_0 h}$ (B) $\frac{3q^2}{4\varepsilon_0 h}$ (C) $\frac{3q}{2\varepsilon_0 h}$ (D) $\frac{3q}{4\varepsilon_0 h}$

38. Monochromatic radiation of wavelength λ is incident on a hydrogen sample containing in ground state. Hydrogen atoms absorb the light and subsequently emit radiations of ten different wavelengths. The value of λ is

(A) 95 nm (B) 103 nm (C) 73 nm (D) 88 nm

39. The electron in hydrogen atom in a sample is in n^{th} excited state, then the number of different spectrum lines obtained in its emission spectrum will be :

(A) 1 + 2 + 3 + ...+ (n – 1) (B) 1 + 2 + 3 + ...+ (n)

(C) 1 + 2 + 3 + ...+ (n + 1) (D) 1 × 2 × 3 × ..... × (n – 1)

40. Consider the spectral line resulting from the transition n = 2 $\rightarrow$ n = 1 in the atoms and ions given below. The shortest wavelength is produced by :

(A) hydrogen atom (B) deuterium atom

(C) singly ionized helium (D) doubly ionized lithium

41. An electron collides with a fixed hydrogen atom in its ground state. Hydrogen atom gets excited and the colliding electron loses all its kinetic energy. Consequently the hydrogen atom may emit a photon corresponding to the largest wavelength of the Balmer series. The min. K.E.of colliding electron will be

(A) 10.2 eV (B) 1.9 eV (C) 12.1 eV (D) 13.6 eV

42. The total energy of a hydrogen atom in its ground state is -13.6eV. If the potential energy in the first excited state is taken as zero then the total energy in the ground state will be :

(A) –3.4eV (B) 3.4 eV (C) – 6.8eV (D) 6.8eV

43. The potential difference applied to an x-ray tube is 5 kV and the current through it is 3.2 mA. Then the number of electrons striking the target per second is

(A) 2×10^{16} (B) 5×10^{6} (C) 1×10^{17} (D) 4×10^{15}

44. The "K_α" X-rays emission line of tungsten occurs at λ = 0.021 nm. The energy difference between K and L levels in this atom is about

(A) 0.51 MeV (B) 1.2 MeV (C) 59 keV (D) 13.6 eV

45. The energy of a photon of characteristic x ray from a Coolidge tube comes from

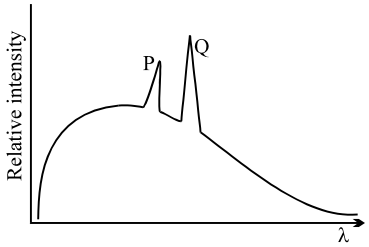
(A) The kinetic energy of the striking electron

(B) The kinetic energy of the free electrons of the target

(C) The kinetic energy of the ions of the target

(D) An atomic transition in the target

46. In a characteristic X-ray spectra of some atom superimposed on continuous X- ray spectra:



(A) P represents K_α line

(B) Q represents K_β line

(C) Q and P represents K_α and K_β lines respectively

(D) Relative positions of K_α and K_β depend on the particular atom

47. The wavelength of the K_α line for an element of atomic number 57 is α. What is the wavelength of the K_α line for the element of atomic number 29 ?

(A) α (B) 2α (C) 4α (D) 8α

48. If the frequencies of $K_{\alpha,}$ K_{β} and L_{α} x rays for a material ν_{K_α}, ν_{K_β} and ν_{L_α}, respectively, then

(A) $\nu_{K_\alpha} = \nu_{K_\beta} + \nu_{L_\alpha}$ (B) $\nu_{L_\alpha} = \nu_{K_\alpha} + \nu_{K_\beta}$ (C) $\nu_{K_\beta} = \nu_{K_\alpha} + \nu_{L_\alpha}$ (D) None of these

49. Consider the nuclear reaction

$$X^{200} \longrightarrow A^{110} + B^{90}$$

If the binding energy per nucleon for X, A and B is 7.4 MeV, 8.2. MeV and 8.2 MeV respectively, what is the energy released ?

(A) 200 MeV (B) 160 MeV (C) 110 MeV (D) 90 MeV

50. The binding energy per nucleon for C^{12} is 7.68 MeV and that for C^{13} is 7.5 MeV. The energy required to remove a neutron from C^{13} is

(A) 5.34 MeV (B) 5.5 MeV (C) 9.5 MeV (D) 9.34 MeV

51. If each fission in a U^{235} nucleus releases 200 MeV, how many fissions must occurs per second to produce a power of 1 KW

(A) 1.325×10^{13} (B) 3.125×10^{13} (C) 1.235×10^{13} (D) 2.135×10^{13}

52. The rest mass of the deuteron, $^{2}_{1}H$, is equivalent to an energy of 1876 MeV, the rest mass of a proton is equivalent to 939 MeV and that of a neutron to 940 MeV. A deuteron may disintegrate to a proton and a neutron if it :

(A) emits a γ-ray photon of energy 2 MeV (B) captures a γ-ray photon of energy 2 MeV
(C) emits a γ-ray photon of energy 3 MeV (D) captures a γ-ray photon of energy 3 MeV

53. In the uranium radioactive series the initial nucleus is $_{92}U^{238}$, and the final nucleus is $_{82}Pb^{206}$. When the uranium nucleus decays to lead, the number of α-particles emitted is.. and the number of β-particles emitted...

(A) 6, 8 (B) 8, 6 (C) 16, 6 (D) 32, 12

54. The radioactive sources A and B of half lifes of 2 hr and 4 hr respectively, initially contain the same number of radioactive atoms. At the end of 2 hours, their rates of disintegration are in the ratio A to B :

(A) 4 : 1 (B) 2 : 1 (C) $\sqrt{2}$: 1 (D) 1 : 1

55. 90% of a radioactive sample is left undecayed after time t has elapsed. What percentage of the initialsample will decay in a total time 2t :

(A) 20% (B) 19% (C) 40% (D) 38%

56. Activity of a radioactive substance is R_1 at time t_1 and R_2 at time $t_2 (t_2 > t_1)$. Then the ratio $\frac{R_2}{R_1}$ is :

(A) $\frac{t_2}{t_1}$ (B) $e^{-\lambda(t_1+t_2)}$ (C) $e\left(\frac{t_1 - t_2}{\lambda}\right)$ (D) $e^{\lambda(t_1-t_2)}$

57. There are two radionuclei A and B. A is an alpha emitter and B is a beta emitter. Their distintegration constants are in the ratio of 1 : 2. What should be the ratio of number of atoms of two at time t = 0 so that probabilities of getting α and β particles are same at time t = 0.

(A) 2 : 1 (B) 1 : 2 (C) e (D) e^{-1}

58. The activity of a sample reduces from A_0 to $A_0/\sqrt{3}$ in one hour. The activity after 3 hours more will be

(A) $\frac{A_0}{3\sqrt{3}}$ (B) $\frac{A_0}{9}$ (C) $\frac{A_0}{9\sqrt{3}}$ (D) $\frac{A_0}{27}$

59. The activity of a sample of radioactive material is A_1 at time t_1 and A_2 at time t_2 ($t_2 > t_1$). Its mean life is T.

(A) $A_1t_1 = A_2t_2$ (B) $\frac{A_1 - A_2}{t_2 - t_1}$ = constant (C) $A_2 = A_1e^{(t_1-t_2)/T}$ (D) $A_2 = A_1e^{(t_1/Tt_2)}$

60. Half life of radium is 1620 years. How many radium nuclei decay in 5 hours in 5 gm radium? (Atomic weight of radium = 223)

(A) 9.1×10^{12} (B) 3.23×10^{15} (C) 1.72×10^{20} (D) 3.3×10^{17}

61. Two radioactive material A_1 and A_2 have decay constants of $10\lambda_0$ and λ_0. If initially they have same number of nuclei, the ratio of number of their undecayed nuclei will be (1/e) after what time

(A) $\frac{1}{\lambda_0}$ (B) $\frac{1}{9\lambda_0}$ (C) $\frac{1}{10\lambda_0}$ (D) 1

62. A certain radioactive nuclide of mass number m_x disintegrates, with the emission of an electron and γ radiation only, to give second nuclied of mass number m_y. Which one of the following equation correctly relates m_x and m_y?

(A) $m_y = m_x + 1$ (B) $m_y = m_x - 2$ (C) $m_y = m_x - 1$ (D) $m_y = m_x$

63. The number of α and β^- emitted during the radioactive decay chain starting from $^{226}_{88}Ra$ and ending at $^{206}_{82}Pb$ is

(A) 3α & $6\beta^-$ (B) 4α & $5\beta^-$ (C) 5α & $4\beta^-$ (D) 6α & $6\beta^-$

64. The decay constant of the end product of a radioactive series is

(A) zero (B) infinite (C) finite (non zero) (D) depends on the end product.

65. At time t = 0, N_1 nuclei of decay constant λ_1 and N_2 nuclei of decay constant λ_2 are mixed. The decay rate of the mixture is :

(A) $N_1N_2e^{-(\lambda_1+\lambda_2)t}$ (B) $+\left(\frac{N_1}{N_2}\right)e^{-(\lambda_1-\lambda_2)t}$

(C) $+\left(N_1\lambda_1e^{-\lambda_1 t} + N_2\lambda_2 e^{-\lambda_2 t}\right)$ (D) $+N_1\lambda_1 N_2\lambda_2 e^{-(\lambda_1+\lambda_2)t}$

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

66. In photoelectric effect, stopping potential depends on

(A) frequency of the incident light (B) intensity of the incident light by varies source distance
(C) emitter's properties (D) frequency and intensity of the incident light

67. Light coming from a discharge tube filled with hydrogen falls on the cathode of the photoelectric cell. The work function of the surface of cathode is 4eV. Which one of the following values of the anode voltage (in Volts) with respect to the cathode will likely to make the photo current zero.

(A) –4 (B) –6 (C) –8 (D) –10

68. An electron in hydrogen atom first jumps from second excited state to first excited state and then, from first excited state to ground state. Let the ratio of wavelength, momentum and energy of photons in the two cases be x, y and z, then select the wrong answer/(s) :
(A) $z = 1/x$ (B) $x = 9/4$ (C) $y = 5/27$ (D) $z = 5/27$

69. Radius of the second Bohr obit of singly ionised helium atom is
(A) 0.53 Å (B) 1.06 Å (C) 0.265 Å (D) 0.132 Å

70. An electron is in an excited state in hydrogen-like atom. It has a total energy of –3.4 eV. If the kinetic energy of the electron is E and its de-Broglie wavelength is λ, then
(A) E = 6.8 eV, $\lambda = 6.6 \times 10^{-10}$ m (B) E = 3.4 eV, $\lambda = 6.6 \times 10^{-10}$ m
(C) E = 3.4 eV, $\lambda = 6.6 \times 10^{-11}$ m (D) E = 6.8 eV, $\lambda = 6.6 \times 10^{-11}$ m

71. An electron in Bohr's hydrogen atom has an energy of –3.4 eV. The angular momentum of the electron is
(A) h / π (B) $h / 2\pi$
(C) $nh / 2\pi$ (n is an integer) (D) $2h / \pi$

72. A beam of ultraviolet light of all wavelengths passes through hydrogen gas at room temperature, in the x-direction. Assume that all photons emitted due to electron transition inside the gas emerge in the y-direction. Let A and B denote the lights emerging from the gas in the x and y directions respectively.
(A) Some of the incident wavelengths will be absent in A.
(B) Only those wavelengths will be present in B which are absent in A.
(C) B will contain some visible light.
(D) B will contain some infrared light.

73. In the hydrogen atom, if the reference level of potential energy is assumed to be zero at the ground state level. Choose the incorrect statement.
(A) The total energy of the shell increases with increase in the value of n
(B) The total energy of the shell decrease with increase in the value of n.
(C) The difference in total energy of any two shells remains the same.
(D) The total energy at the ground state becomes 13.6 eV.

74. Let A_n be the area enclosed by the nth orbit in a hydrogen atom. The graph of $\ln (A_n/A_1)$ agains ln (n).
(A) will pass through origin (B) will be a stright line with slope 4
(C) will be a monotonically increasing nonlinear curve (D) will be a circle.

75. An electron in hydrogen atom after absorbing energy photons can jump between energy states n_1 and $n_2 (n_2 > n_1)$. Then it may return to ground state after emitting six different wavelengths in emission spectrum. the energy of emitted photons is equal to, less than and greater than the absorbed photons. Then n_1 and n_2 are :
(A) $n_2 = 4, n_1 = 3$ (B) $n_2 = 5, n_1 = 3$ (C) $n_2 = 4, n_1 = 2$ (D) $n_2 = 4, n_1 = 1$

76. A neutron collides head on with a stationary hydrogen atom in ground state
(A) If kinetic energy of the neutron is less than 13.6 eV, collision must be elastic
(B) if kinetic energy of the neutron is less than 13.6 eV, collision may be inelastic.
(C) inelastic collision takes place when initial kinetic energy of neutron is greater than 13.6 eV.
(D) perfectly inelastic collision cannot take place.

77. A neutron collides head-on with a stationary hydrogen atom in ground state. Which of the following statements are correct (Assume that the hydrogen atom and neutron has same mass) :
(A) If kinetic energy of the neutron is less than 20.4 eV collision must be elastic.
(B) If kinetic energy of the neutron is less than 20.4 eV collision may be inelastic.
(C) Inelastic collision may be take place only when initial kinetic energy of neutron is greater than 20.4 eV.
(D) Perfectly inelastic collision can not take place.

78. When a nucleus with atomic number Z and mass number A undergoes a radioactive decay process:
(A) both Z and A will decrease, if the process is α decay
(B) Z will decrease but A will not change, if the process is β^+ decay
(C) Z will decrease but A will not change, if the process is β^- decay
(D) Z and A will remain unchanged, if the process is γ decay.

79. de-Broglie wavelength of an electron in the nth Bohr orbit is λ_n and the angular momentum is J_n, then:

(A) $J_n \propto \lambda_n$ (B) $\lambda_n \propto \frac{1}{J_n}$ (C) $\lambda_n \propto J_n^2$ (D) none of these

80. In a Coolidge tube experiment, the minimum wavelength of the continuous X-ray spectrum is equal to 66.3 pm, then
(A) electrons accelerate through a potential difference of 12.75 kV in the Coolidge tube
(B) electrons accelerate through a potential difference of 18.75 kV in the Coolidge tube
(C) de-Broglie wavelength of the electrons reaching the anticathode is of the order of 10 μm.
(D) de-Broglie wavelength of the electrons reaching the anticathode is 0.01Å.

81. The potential difference applied to an X-ray tube is increased. As a result, in the emitted radiation :
(A) the intensity increases (B) the minimum wave length increases
(C) the intensity decreases (D) the minimum wave length decreases

82. An x-ray tube operates at an accelerating potential of 20 kV. Which of the following wavelengths will be absent in the continuous spectrum of x ray ?
(A) 12 pm (B) 45 pm (C) 65 pm (D) 95 pm

83. The K, L and M energy levels of platinum lie roughly at 78, 12 and 3 KeV, respectively. The ratio of wavelength of K_α line to that of K_β line in x-ray spectrum is

(A) $\frac{22}{3}$ (B) $\frac{3}{22}$ (C) $\frac{22}{25}$ (D) $\frac{25}{22}$

84. The energy of a tungsten atom with a vacancy in L shell is 11.3 keV. Wavelength of K_α photon for tungsten is 21.3 pm. If a potential difference of 62 kV is applied across the x-rays tube following characteristic x-rays will be produced.
(A) K, L series (B) Only K_α and L series
(C) Only L series (D) None of these

85. When the atomic number A of the nucleus increases
(A) initially the neutron-proton ratio is constant = 1
(B) initially neutron-proton ratio increases and later decreases
(C) initially binding energy per nucleon increases and later decreases
(D) the binding energy per nucleon increases when the neutron-proton ratio increases.

86. Let m_p be the mass of a proton, m_n the mass of a neutron, M_1 the mas of a $^{20}_{10}\mathrm{Ne}$ nucleus and M_2 the mass of a $^{40}_{20}\mathrm{Ca}$ nucleus. Then
(A) $M_2 = 2M_1$ (B) $M_2 > 2M_1$ (C) $M_2 < 2M_1$ (D) $M_1 < 10(m_n + m_p)$

87. In an α-decay the Kinetic energy of α particle is 48 MeV and Q-value of the reaction is 50 MeV. The mass number of the mother nucleus is: (Assume that daughter nucleus is in ground state)
(A) 96 (B) 100 (C) 104 (D) none of these

88. A radioactive material of half-life T was produced in a nuclear reactor at different instants, the quantity produced second time was twice of that produced first time. If now their present activities are A_1 and A_2 respectively then their age difference equals :

(A) $\frac{T}{\ln 2}\left|\ln\frac{A_1}{A_2}\right|$ (B) $T\left|\ln\frac{A_1}{A_2}\right|$ (C) $\frac{T}{\ln 2}\left|\ln\frac{A_2}{2A_1}\right|$ (D) $T\left|\ln\frac{A_2}{2A_1}\right|$

89. Half life for certain radioactive element is 5 min. Four nuclei of that element are observed at a certain instant of time. After five minutes

Assertion (A) : It can be definitely said that two nuclei will be left undecayed.

Reasoning (R) : After half life i.e. 5 minutes, half of total nuclei will disintegrate. So only two nuclei will be left undecayed. Then

(A) **A** is correct & **R** is correct explanation of **A**.

(B) Both are correct. But **R** is not correct explanation of **A**.

(C) **A** is incorrect & **R** is correct.

(D) Both are incorrect.

90. In the following, **column-I** lists some physical quantities & the **column-II** gives approx. energy values associated with some of them. Choose the appropriate value of energy from **column-II** for each of the physical quantities in **column-I** and write the corresponding letter A, B, C etc. against the number (i), (ii), (iii), etc. of the physical quantity in the answer book. In your answer, the sequence of **column-I** should be maintained.

Column-I	**Column-II**
(i) Energy of thermal neutrons	(A) 0.025 eV
(ii) Energy of X-rays	(B) 0.5 eV
(iii) Binding energy per nucleon	(C) 3 eV
(iv) Photoelectric threshold of metal	(D) 20 eV
	(E) 10 keV
	(F) 8 MeV

EXERCISE-II

1. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy T_a eV and de Broglie wavelength λ_a. The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.7 eV is $T_b = (T_a - 1.5)$ eV. If the De Broglie wavelength of these photoelectrons is $\lambda_b = 2\lambda_a$, then find
(a) The work function of a (b) The work function of b is (c) T_a and T_b

2. When a monochromatic point source of light is at a distance of 0.2 m from a photoelectric cell, the cut off voltage and the saturation current are respectively 0.6 volt and 18.0 mA. If the same source is placed 0.6 m away from the photoelectric cell, then find
(a) the stopping potential (b) the saturation current

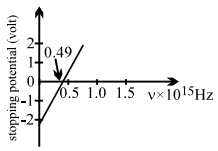
3. An isolated metal body is illuminated with monochromatic light and is observed to become charged to a steady positive potential 1.0 V with respect to the surrounding. The work function of the metal is 3.0 eV. The frequency of the incident light is _____.

4. 663 mW of light from a 540 nm source is incident on the surface of a metal. If only 1 of each 5×10^9 incident photons is absorbed and causes an electron to be ejected from the surface, the total photocurrent in the circuit is _____.

5. Light of wavelength 330 nm falling on a piece of metal ejects electrons with sufficient energy which requires voltage V_0 to prevent a electron from reaching collector. In the same setup, light of wavelength 220 nm, ejects electrons which require twice the voltage V_0 to stop them in reaching a collector. Find the numerical value of voltage V_0. (Take plank's constant, $h = 6.6 \times 10^{-34}$ Js and 1 eV = 1.6×10^{-19} J)

6. A small 10W source of ultraviolet light of wavelength 99 nm is held at a distance 0.1 m from a metal surface. The radius of an atom of the metal is approximately 0.05 nm. Find
(i) the average number of photons striking an atom per second.
(ii) the number of photoelectrons emitted per unit area per second if the efficiency of liberation of photoelectrons is 1%.

7. The surface of cesium is illuminated with monochromatic light of various wavelengths and the stopping potentials for the wavelengths are measured. The results of this experiment is plotted as shown in the figure. Estimate the value of work function of the cesium and Planck's constant.



8. A parallel beam of uniform, monochromatic light of wavelength 2640 A has an intensity of 200 W/m^2. The number of photons in 1mm^3 of this radiation are .....

9. An electron of mass "m" and charge "e" initially at rest gets accelerated by a constant electric field E. The rate of change of De-Broglie wavelength of this electron at time t is .....

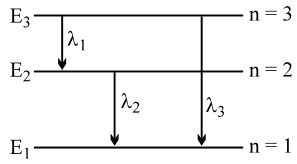
10. A hydrogen atom in a state having a binding energy 0.85 eV makes a transition to a state of excitation energy 10.2 eV. The wave length of emitted photon is .....nm.

11. A hydrogen atom is in 5th excited state. When the electron jumps to ground state the velocity of recoiling hydrogen atom is ..... m/s and the energy of the photon is .....eV.

12. The ratio of series limit wavlength of Balmer series to wavelength of first line of paschen series is .....

13. An electron joins a helium nucleus to form a He^+ ion in ground state. The wavelength of the photon emitted in this process if the electron is assumed to have had no kinetic energy when it combines with nucleus is .....nm.

14. Three energy levels of an atom are shown in the figure. The wavelength corresponding to three possible transition are λ_1, λ_2 and λ_3.
The value of λ_3 in terms of λ_1 and λ_2 is given by _____.



15. Imagine an atom made up of a proton and a hypothetical particle of double the mass of an electron but having the same charge as the electron. Apply the Bohr atom model and consider a possible transitions of this hypothetical particle to the first excited level. Find the longest wavelngth photon that will be emitted λ (in terms of the Rydberg constant R.)

16. In a hydrogen atom, the electron moves in an orbit of radius 0.5 Å making 10^{16} revolution per second. The magnetic moment associated with the orbital motion of the electron is _____.

17. A hydrogen like atom has its single electron orbiting around its stationary nucleus. The energy to excite the electron from the second Bohr orbit to the third Bohr orbit is 47.2 eV. The atomic number of this nucleus is_____.

18. A single electron orbits a stationary nucleus of charge Ze where Z is a constant and e is the electronic charge. It requires 47.2 eV to excite the electron from the 2nd Bohr orbit to 3rd Bohr orbit. Find
(i) the value of Z,
(ii) energy required to excite the electron from the third to the fourth orbit
(iii) the wavelength of radiation required to remove the electron from the first orbit to infinity
(iv) the kinetic energy, potential energy and angular momentum in the first Bohr orbit
(v) the radius of the first Bohr orbit.

19. A hydrogen like atom (atomic number Z) is in higher excited state of quantum number n. This excited atom can make a transition to the first excited state by successively emitting two photons of energy 22.95 eV and 5.15 eV respectively. Alternatively, the atom from the same excited state can make transition to the second excited state by successively emitting two photons of energies 2.4 eV and 8.7 eV respectively. Find the values of n and Z.

20. Find the binding energy of an electron in the ground state of a hydrogen like atom in whose spectrum the third of the corresponding Balmer series is equal to 108.5nm.

21. Which level of the doubly ionized lithium has the same energy as the ground state energy of the hydrogen atom. Find the ratio of the two radii of corresponding orbits.

22. The binding energies per nucleon for deuteron ($_1H^2$) and helium ($_2He^4$) are 1.1 MeV and 7.0 MeV respectively. The energy released when two deuterons fuse to form a helium nucleus ($_2He^4$) is _____.

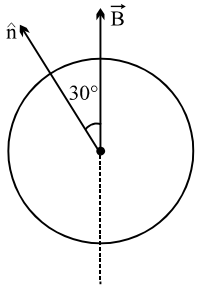
23. A radioactive decay counter is switched on at t = 0. A β-active sample is present near the counter. The counter registers the number of β-particles emitted by the sample. The counter registers 1×10^5 β-particles at t = 36 s and 1.11×10^5 β-particles at t = 108 s. Find $T_{1/2}$ of this sample

24. At t = 0, a sample is placed in a reactor. An unstable nuclide is produced at a constant rate R in the sample by neutron absorption. This nuclide β^- decays with half life τ. Find the time required to produce 80% of the equilibrium quantity of this unstable nuclide.

25. Suppose that the Sun consists entirely of hydrogen atom and releases the energy by the nuclear reaction, $4\ {}^1_1H \rightarrow {}^4_2He$ with 26 MeV of energy released. If the total output power of the Sun is assumed to remain constant at 3.9×10^{26} W, find the time it will take to burn all the hydrogen. Take the mass of the Sun as 1.7×10^{30} kg.

26. The positron is a fundamental particle with the same mass as that of the electron and with a charge equal to that of an electron but of opposite sign. When a positron and an electron collide, they may annihilate each other. The energy corresponding to their mass appears in two photons of equal energy. Find the wavelength of the radiation emitted.
[Take : mass of electron = $(0.5/C^2)$MeV and hC = 1.2×10^{-12} MeV.m where h is the Plank's constant and C is the velocity of light in air]

27. An electron in the ground state of hydrogen atoms is revolving in anti-clockwise direction in a circular orbit of radius R.



(i) Obtain an expression for the orbital magnetic dipole moment of the electron.
(ii) The atom is placed in a uniform magnetic induction, such that the plane normal to the electron orbit make an angle of 30º with the magnetic induction. Find the torque experienced by the orbiting electron.

28. A small plate of a metal (work function = 1.17 eV) is placed at a distance of 2m from a monochromatic light source of wavelength 4.8×10^{-7} m and power 1.0 watt. The light falls normally on the plate. Find the number of photons striking the metal plate per square meter per sec. If a constant uniform magnetic field of strength 10^{-4} tesla is applied parallel to the metal surface. Find the radius of the largest circular path followed by the emitted photoelectrons.

29. A beam of light has three wavelengths 4144Å, 4972Å & 6216Å with a total intensity of 3.6×10^{-3} W.m^{-2} equally distributed amongst the three wavelengths. The beam falls normally on an area 1.0 cm^2 of a clean metallic surface of work function 2.3 eV. Assume that there is no loss of light by reflection and that each energetically capable photon ejects one electron. Calculate the number of photoelectrons liberated in two seconds.

30. A neutron with kinetic energy 25 eV strikes a stationary deuteron. Find the de Broglie wavelengths of both particles in the frame of their centre of mass before collision.

31. Two identical nonrelativistic particles move at right angles to each other, possessing De Broglie wavelengths, λ_1 and λ_2. Find the De Broglie wavelength of each particle in the frame of their centre of mass.

32. A stationary He^+ ion emitted a photon corresponding to the first line its Lyman series. That photon liberated a photoelectron from a stationary hydrogen atom in the ground state. Find the velocity of the photoelectron.

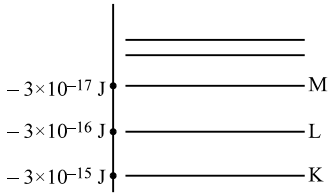
33. A gas of identical hydrogen like atoms has some atoms in the lowest (ground) energy level A & some atoms in a particular upper (excited) energy level B & there are no atoms in any other energy level. The atoms of the gas make transition to a higher energy level by the absorbing monochromatic light of photon energy 2.7eV. Subsequently, the atoms emit radiation of only six different photon energies. Some of the emitted photons have energy 2.7 eV. Some have energy more and some have less than 2.7 eV.
(i) Find the principal quantum number of the initially excited level B.
(ii) Find the ionisation energy for the gas atoms.
(iii) Find the maximum and the minimum energies of the emitted photons.

34. A hydrogen atom in ground state absorbs a photon of ultraviolet radiation of wavelength 50 nm. Assuming that the entire photon energy is taken up by the electron, with what kinetic energy will the electron be ejected?

35. A monochromatic light source of frequency ν illuminates a metallic surface and ejects photoelectrons. The photoelectrons having maximum energy are just able to ionize the hydrogen atoms in ground state. When the whole experiment is repeated with an incident radiation of frequency $(5/6)\nu$, the photoelectrons so emitted are able to excite the hydrogen atom beam which then emits a radiation of wavelength of 1215 Å. Find the work function of the metal and the frequency ν.

36. An energy of 68.0 eV is required to excite a hydrogen like atom from its second Bohr orbit to the third. The nuclear charge Ze. Find the value of Z, the kinetic energy of the electron in the first Bohr orbit and the wavelength of the electro magnetic radiation required to eject the electron from the first Bohr orbit to infinity.

37. Simplified picture of electron energy levels in a certain atom is shown in the figure. The atom is bombarded with high energy electrons. The impact of one of these electron has caused the complete removal of K-level is filled by an electron from the L-level with a certain amount of energy being released during the transition. This energy may appear as X-ray or may all be used to eject an M-level electron from the atom.



Find :

(i) the minimum potential difference through which electron may be accelerated from rest to cause the ejectrion of K-level electron from the atom.

(ii) energy released when L-level electron moves to fill the vacancy in the K-level.

(iii) wavelength of the X-ray emitted.

(iv) K.E. of the electron emitted from the M-level.

38. U^{238} and U^{235} occur in nature in an atomic ratio 140 : 1. Assuming that at the time of earth's formation the two isotopes were present in equal amounts. Calculate the age of the earth.
(Half life of $u^{238} = 4.5 \times 10^9$ yrs & that of $U^{235} = 7.13 \times 10^8$ yrs)

39. The kinetic energy of an $\alpha-$particle which flies out of the nucleus of a Ra^{226} atom in radioactive disintegration is 4.78 MeV. Find the total energy evolved during the escape of the α-particle.

40. A small bottle contains powdered beryllium Be & gaseous radon which is used as a source of α-particles. Neutrons are produced when α-particles of the radon react with beryllium. The yield of this reaction is (1/ 4000) i.e. only one α-particle out of 4000 induces the reaction. Find the amount of radon (Rn^{222}) originally introduced into the source, if it produces 1.2×10^6 neutrons per second after 7.6 days.
[$T_{1/2}$ of R_n = 3.8 days]

41. A wooden piece of great antiquity weighs 50 gm and shows C^{14} activity of 320 disintegrations per minute. Estimate the length of the time which has elapsed since this wood was part of living tree, assuming that living plants show a C^{14} activity of 12 disintegrations per minute per gm. The half life of C^{14} is 5730 yrs.

42. A radionuclide with disintegration constant λ is produced in a reactor at a constant rate α nuclei per sec. During each decay energy E_0 is released. 20% of this energy is utilised in increasing the temperature of water. Find the increase in temperature of m mass of water in time t. Specific heat of water is S. Assume that there is no loss of energy through water surface.

43. In a photo electric effect set-up, a point source of light of power 3.2×10^{-3} W emits mono energetic photons of energy 5.0 eV. The source is located at a distance of 0.8 m from the centre of a stationary metallic sphere of work function 3.0 eV & of radius 8.0×10^{-3} m. The efficiency of photo electrons emission is one for every 10^6 incident photons. Assume that the sphere is isolated and initially neutral, and that photo electrons are instantly swept away after emission.

(a) Calculate the number of photo electrons emitted per second.

(b) Find the ratio of the wavelength of incident light to the De-Broglie wave length of the fastest photo electrons emitted.

(c) It is observed that the photo electron emission stops at a certain time t after the light source is switched on. Why?

(d) Evaluate the time t.

44. Assume that the de-Broglie wave associated with an electron can form a standing wave between the atoms arranged in a one dimensional array with nodes at each of the atomic sites. It is found that one such standing wave is formed if the distance 'd' between the atoms of the array is 2 Å. A similar standing wave is again formed if 'd' is increased to 2.5 Å but not for any intermediate value of d. Find the energy of the electrons in electron volts and the least value of d for which the standing wave of the type described above can form.

45. The element Curium $^{248}_{96}\mathrm{Cm}$ has a mean life of 10^{13} seconds. Its primary decay modes are spontaneous fission and α decay, the former with a probability of 8% and the latter with a probability of 92%. Each fission releases 200 MeV of energy. The masses involved in α decay are as follows :

$^{248}_{96}\mathrm{Cm} = 248.072220\,\mathrm{u}$, $^{244}_{94}\mathrm{Pu} = 244.064100\,\mathrm{u}$ & $^{4}_{2}\mathrm{He} = 4.002603\,\mathrm{u}$.

Calculate the power output from a sample of 10^{20} Cm atoms. (1 u = 931 MeV/c^2)

EXERCISE-III

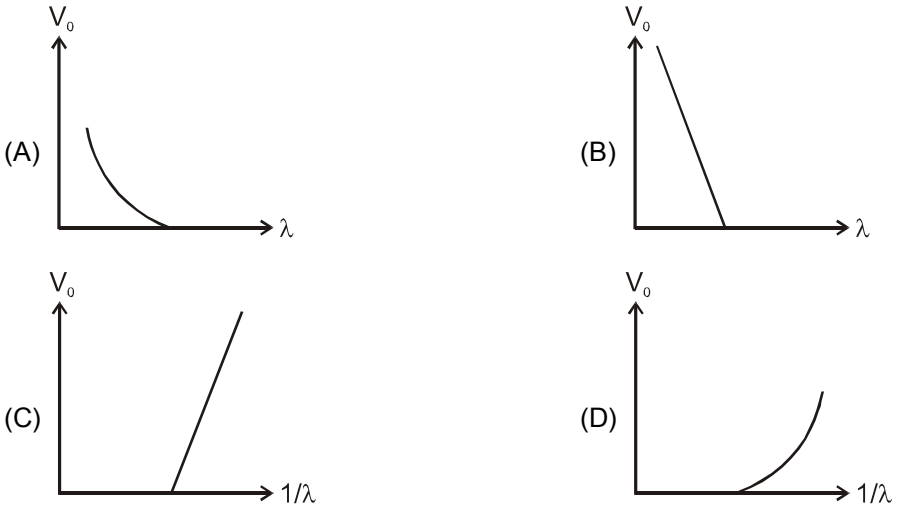
1. A metal surface is illuminated by light of two different wavelengths 248 *nm* and 310 *nm*. The maximum speeds of the photoelectrons corresponding to these wavelengths are u_1 and u_2, respectively. If the ratio $u_1 : u_2 = 2 : 1$ and $hc = 1240\ eV\ nm$, the work function of the metal is nearly **[JEE Advanced- 2014]**

(A) 3.7 *eV* (B) 3.2 *eV* (C) 2.8 *eV* (D) 2.5 *eV*

2. If l_{Cu} is the wavelength of K_a X-ray line of copper (atomic number 29) and λ_{Mo} is the wavelength of the K_a X-ray line of molybdenum (atomic number 42), then the ratio $\lambda_{Cu} / \lambda_{Mo}$ is close to **[JEE Advanced- 2014]**

(A) 1.99 (B) 2.14 (C) 0.50 (D) 0.48

3. For photo-eledctric effect with incident photon wavelength λ, the stopping potential is V_0. Identify the correct variation(s) of V_0 with λ and $1/\lambda$. **[JEE Advanced- 2015]**



4. A fission reaction is given by ${}^{236}_{92}U \rightarrow {}^{140}_{54}Xe + {}^{94}_{38}Xe + {}^{94}_{38}Sr + x + y$, where x and y are two particles. Considering ${}^{236}_{92}U$ to be at rest, the kinetic energies of the products are denoted by K_{Xe}, K_{Sr}, K_x(2MeV) and K_y(2 MeV), respectively. Let the binding energies per nucleon of ${}^{236}_{92}U$, ${}^{140}_{54}Xe$ and ${}^{94}_{38}Sr$ be 7.5 MeV, 8.5 MeV and 8.5 MeV, respectively. Considering different conservation laws, the correct option(s) is(are)

[JEE Advanced- 2015]

(A) x = n, y = n, K_{Sr} = 129 MeV, K_{Xe} = 86 MeV
(B) x = p, y = e^-, K_{Sr} = 129 MeV, K_{Xe} = 86 MeV
(C) x = p, y = n, K_{Sr} = 129 MeV, K_{Xe} = 86 MeV
(D) x = n, y = n, K_{Sr} = 86 MeV, K_{Xe} = 129 MeV

5. Match the nuclear processes given in column-I with the appropriate option(s) in column-II **[JEE Advanced- 2015]**

Column-I	**Column-II**
(A) Nuclear fusion	(P) Absorption of thermal neutrons by ${}^{235}_{92}U$
(B) Fission in a nuclear reactor	(Q) ${}^{60}_{27}Co$ nucleus
(C) β-decay	(R) Energy production in stars via hydrogen conversion to helium
(D) γ-ray emission	(S) Heavy water
	(T) Neutrino emission

6. Consider a hydrogen atom with its electron in the nth orbital. An electromagnetic radiation of wavelength 90 nm is used to ionize the atom. If the kinetic energy of the ejected electron is 10.4eV, then the value of n is (hc = 1242 eV nm) **[JEE Advanced- 2015]**

7. A nuclear power plant supplying electrical power to a village uses a radioactive material of half life T years as the fuel. The amount of fuel at the beginning is such that the total power requirement of the village is 12.5% of the electrical power available from the plant at that time. If the plant is able to meet the total power needs of the village for a maximum period of nT years, then the value of n is : **[JEE Advanced- 2015]**

8. For a radioactive material, its activity A and rate of change of its activity R are defined as $A = -\frac{dN}{dt}$ and $R = -\frac{dA}{dt}$, where N(t) is the number of nuclei at time t. Two radioactive sources P(mean life τ) and Q(mean life 2τ) have the same activity at t = 0. Their rates of change of activities at t = 2τ are R_P and R_Q, respectively. If $\frac{R_P}{R_Q} = \frac{n}{e}$, then the value of n is : **[JEE Advanced- 2015]**

9. An electron in an excited state of Li^{2+} ion has angular momentum $3h/2\pi$. The de-Broglie wavelength of the electron in this state is $p\pi a_0$ (where a_0 is the Bohr radius). The value of p is : **[JEE Advanced-2015]**

10. In a historical experiment to determine Planck's constant, a metal surface was irradiated with light of different wavelengths. The emitted photoelectron energies were measured by applying a stopping potential. The relevant data for the wavelength (λ) of incident light and the corresponding stopping potential (V_0) are given below :

$\lambda(\mu m)$	V_0(Volt)
0.3	2.0
0.4	1.0
0.5	0.4

Given that c= 3×10^8 m s^{-1} and e = 1.6×10^{-19} C, Planck's constant (in units of J s) found from such an experiment is **[JEE Advanced- 2016]**

(A) 6.0×10^{-34} (B) 6.4×10^{-34} (C) 6.6×10^{-34} (D) 6.8×10^{-34}

11. Highly excited states for hydrogen-like atoms (also called Rydberg states) with nuclear charge Ze are defined by their principal quantum number n, where n >> 1. Which of the following statement(s) is(are) true? **[JEE Advanced-2016]**

(A) Relative change in the radii of two consecutive orbitals does not depends on Z

(B) Relative change in the radii of two consecutive orbitals varies as 1/n

(C) Relative change in the energy of two consecutive orbitals varies as $1/n^3$

(D) Relative change in the angular momenta of two consecutive orbitals varies as 1/n

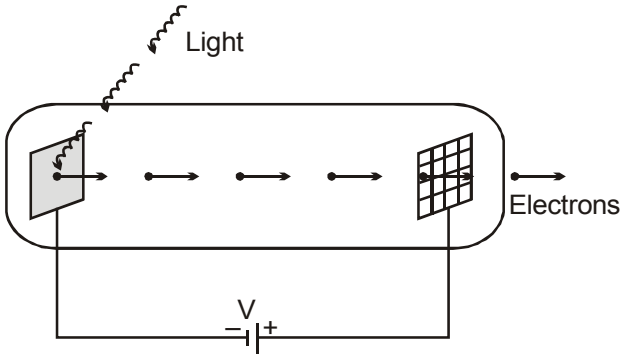
12. The electrostatic energy of Z protons uniformly distributed throughout a spherical nucleus of radius R is given by **[JEE Advanced-2016]**

$$E = \frac{3}{5}\frac{Z(Z-1)e^2}{4\pi\varepsilon_0 R}$$

The measured masses of the neutron, ${}_1^1H$, ${}_7^{15}N$ and ${}_8^{15}O$ are 1.008665 u, 1.007825 u, 15.000109 u and 15.003065 u, respectively. Given that the radii of both the ${}_7^{15}N$ and ${}_8^{15}O$ nuclei are same, 1 u = 931.5 MeV/c^2 (c is the speed of light) and $e^2/(4\pi\varepsilon_0)$ = 1.44 MeV fm. Assuming that the difference between the binding energies of ${}_7^{15}N$ and ${}_8^{15}O$ is purely due to the electrostatic energy, the radius of either of the nuclei is (1 fm=10^{-15} m)

(A) 2.85 fm (B) 3.03 fm (C) 3.42 fm (D) 3.80 fm

13. Light of wavelength λ_{ph} falls on a cathode plate inside a vacuum tube as shown in the figure. The work function of the cathode surface is ϕ and the anode is a wire mesh of conducting material kept at a distance d from the cathode. A potential difference V is maintained between the electrodes. If the minimum de Broglie wavelength of the electrons passing through the anode is λ_e, which of the following statement(s) is(are) true? **[JEE Advanced-2016]**



(A) For large potential difference (V >> ϕ/e), λ_e is approximately halved if V is made four times.
(B) λ_e increases at the same rate as λ_{ph} for $\lambda_{ph} < hc/\phi$
(C) λ_e is approximately halved, if d is doubled.
(D) λ_e decreases with increase in ϕ and λ_{ph}

14. A hydrogen atom in its ground state is irradiated by light of wavelength 970 Å. Taking hc/e = 1.237 × 10^{-6} eV m and the ground state energy of hydrogen atom at –13.6 eV, the number of lines present in the emission spectrum is : **[JEE Advanced-2016]**

15. The isotope ${}_5^{12}B$ having a mass 12.014 u undergoes β-decay to ${}_6^{12}C$. ${}_6^{12}C$ has an excited state of the nucleus (${}_6^{12}C^*$) at 4.041 MeV above its ground state. If ${}_5^{12}B$ decays to ${}_6^{12}C^*$, the maximum kinetic energy of the β-particle in units of MeV is
(1 u = 931.5 MeV/c^2, where c is the speed of light in vacuum). **[JEE Advanced-2016]**

16. A photoelectric material having work-function ϕ_0 is illuminated with light of wavelength $\lambda\left(\lambda < \frac{hc}{\phi_0}\right)$. The fastest photoelectron has a de Broglie wavelength λ_d. A change in wavelength of the incident light by $\Delta\lambda$ results in a change $\Delta\lambda_d$ in λ_d. Then the ratio $\Delta\lambda_d/\Delta\lambda$ is proportional to **[IIT-Advanced-2017]**

(A) $\frac{\lambda_d^3}{\lambda^2}$ (B) $\frac{\lambda_d^3}{\lambda}$ (C) $\frac{\lambda_d^2}{\lambda^2}$ (D) $\frac{\lambda_d}{\lambda}$

17. ^{131}I is isotope of iodine that β decays to an isotope of Xenon with a half-life of 8 days. A small amount of a secum labelled with ^{131}I is injected into the blood of a person. The activity of the amound of ^{131}I injected was 2.4×10^5 Becquerel (Bq). It is known that the injected serum will get distributed uniformly in the blood stream in less than half an hour. After 11.5 hours, 2.5 ml of blood is drawn from the person's body, and gives an activity of 115 Bq. The total volume of blood in the person's body, in liters is approximately (you may use $e^x \approx 1 + x$ for $|x| << 1$ and $\ln 2 \approx 0.7$). **[IIT-Advanced-2017]**

18. An electron in a hydrogen atom undergoes a transition from an orbit with quantum number n_i to another with quantum number n_f. V_i and V_f are respectively the initial and final potential energies of the electron. If $\frac{V_i}{V_f} = 6.25$, then the smallest posssible n_f is : **[IIT-Advanced-2017]**

19. In a radioactive decay chain, $^{232}_{90}Th$ nucleus decays to $^{212}_{82}Pb$ nucleus. Let N_α and N_β be the number of α and β^- particles, respectively, emitted in this decay process. Which of the following statements is(are) true? **[IIT-Advanced-2018]**

(A) $N_\alpha = 5$ (B) $N_\alpha = 6$ (C) $N_\beta = 2$ (D) $N_\beta = 4$

20. In a photoelectric experiment a parallel beam of monochromatic light with power of 200 W is incident on a perfectly absorbing cathode of work function 6.25 eV. The frequency of light is just above the threshold frequency so that the photoelectrons are emitted with negligible kinetic energy. Assume that the photoelectron emission efficiency is 100%. A potential difference of 500 V is applied between the cathode and the anode. All the emitted electrons are incident normally on the anode and are absorbed. The anode experiences a force $F = n \times 10^{-4}$ N due to the impact of the electrons. The value of n is _____.
Mass of the electrons $m_e = 9 \times 10^{-31}$ kg and 1.0 eV = 1.6×10^{-19} J. **[IIT-Advanced-2018]**

21. Consider a hydrogen-like ionized atom with atomic number Z with a single electron. In the emission spectrum of this atom, the photon emitted in the n = 2 to n = 1 transition has energy 74.8 eV higher than the photon emitted in the n = 3 to n = 2 transition. The ionization energy of the hydrogen atom is 13.6 eV. The value of Z is _____. **[IIT-Advanced-2018]**

22. In a radioactive sample $^{40}_{19}K$ nuclei with decay into stable $^{40}_{20}Ca$ nuclei with decay constant 4.5×10^{-10} per year or into stable $^{40}_{18}Ar$ nuclei with decay constant 0.5×10^{-10} per year. Given that in this sample all the stable $^{40}_{20}Ca$ and $^{40}_{18}Ar$ nuclei are produced by the $^{40}_{19}K$ nuclei only. In time $t \times 10^9$ years, if the ratio of the sum of stable $^{40}_{20}Ca$ and $^{40}_{18}Ar$ nuclei to the radioactive $^{40}_{19}K$ nuclei is 99, the value of t will be.

[Given : ln 10 = 2.3] **[IIT-Advanced-2019]**

(A) 1.15 (B) 2.3 (C) 9.2 (D) 4.6

23. A free hydrogen atom after absorbing a photon of wavelength λ_a gets excited from the state n = 1 to the state n = 4. Immediately after that the electron jumps to n = m state by emitting a photon of wavelength λ_e. Let the change in momentum of atom due to the absorption and the emission are Δp_a and Δp_e, respectively. If $\frac{\lambda_a}{\lambda_e} = \frac{1}{5}$, which of the option(s) is/are correct ? **[IIT-Advanced-2019]**

[Use hc = 1242 eV nm ; 1 nm = 10^{-9} m, h and c are Planck's constant and speed of light, respectively]

(A) The ratio of kinetic energy of the electron in the state n = m to the state n = 1 is $\frac{1}{4}$

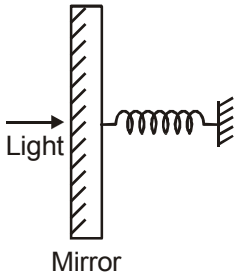
(B) $\frac{\Delta p_a}{\Delta p_e} = \frac{1}{2}$ (C) m = 2 (D) $\lambda_e = 418$ nm

24. Suppose a $^{226}_{88}Ra$ nucleus at rest and in ground state undergoes α-decay to a $^{222}_{86}Rn$ nucleus in its excited state. The kinetic energy of the emitted α-particle is found to be 4.44 MeV. $^{222}_{86}Rn$ nucleus then goes to its ground state by γ-decay. The energy of the emitted γ photon is _____ keV.

[Given : atomic mass of $^{226}_{88}Ra$ = 226.005 u, atomic mass of $^{222}_{86}Rn$ = 222.000 u, atomic mass of α-particle = 4.000 u, 1 u = 931 MeV./c^2 . c is speed of the light] **[IIT-Advanced-2019]**

25. A perfectly reflecting mirror of mass M mounted on a spring constitutes a spring-mass system of angular frequency Ω such that $\frac{4\pi m\Omega}{h} = 10^{24}\,m^{-2}$ with h as Planck's constant. N photons of wavelength $\lambda = 8\pi \times 10^{-6}$ m strike the mirror simultaneously at normal incidence such that the mirror gets displaced by 1 μm. If the value of N is $x \times 10^{12}$, then the value of x is _____.

[Consider the spring as massless] **[IIT-Advanced-2019]**



26. A particle of mass m moves in circular orbits with potential energy V(r) = Fr, where F is a positive constant and r is its distance from the origin. Its energies are calculated using the Bohr model. If the radius of the particle's orbit is denoted by R and its speed and energy are denoted by v and E, respectively, then for the nth orbit (here h is the Planck's constant) **[IIT-Advanced-2020]**

(A) $R \propto n^{1/3}$ and $v \propto n^{2/3}$ (B) $R \propto n^{2/3}$ and $v \propto n^{1/3}$ (C) $E = \frac{3}{2}\left(\frac{n^2h^2F^2}{4\pi^2m}\right)^{1/3}$ (D) $E = 2\left(\frac{n^2h^2F^2}{4\pi^2m}\right)^{1/3}$

27. In an X-ray tube, electrons emitted from a filament (cathode) carrying current I hit a target (anode) at a distance d from the cathode. The target is kept at a potential V higher than the cathode resulting in emission of continuous and characteristic X-rays. If the filament current I is decreased to $\frac{I}{2}$, the potential difference V is increased to 2V, and the separation distance d is reduced to $\frac{d}{2}$, then

(A) the cut-off wavelength will reduce to half, and the wavelengths of the characteristic X-rays will remain the same

(B) the cut-off wavelength as well as the wavelengths of the characteristic X-rays will remain the same

(C) the cut-off wavelength will reduce to half, and the intensities of all the X-rays will decrease

(D) the cut-off wavelength will become two times larger, and the intensity of all the X-rays will decrease

[IIT-Advanced-2020]

28. A particular hydrogen like ion emits radiation of frequency 2.92×10^{15} Hz when it makes transition from n = 3 to n = 1. The frequency in Hz of radiation emitted in transition from n = 2 to n = 1 will be :

[IIT-Main-2021]

(A) 6.57×10^{15} (B) 4.38×10^{15} (C) 0.44×10^{15} (D) 2.46×10^{15}

29. A free electron of 2.6 eV energy collides with a H^+ ion. This results in the formation of a hydrogen atom in the first excited state and a photon is released. Find the frequency of the emitted photon. ($h = 6.6 \times 10^{-34}$J] s):

[IIT-Main-2021]

(A) 1.45×10^9 MHz (B) 9.0×10^{27} MHz (C) 1.45×10^{16} MHz (D) 0.19×10^{15} MHz

30. The de-Broglie wavelength of a particle having kinetic energy E is λ. How much extra energy must be given to this particle so that the de-Broglie wavelength reduces to 75% of the initial value ?**[IIT-Main-2021]**

(A) E (B) $\frac{7}{9}E$ (C) $\frac{1}{9}E$ (D) $\frac{16}{9}E$

31. The temperature of an ideal gas in 3-dimensions is 300 K. The corresponding de-Broglie wavelength of the electron approximately at 300 K, is :

[m_e = mass of electron = 9×10^{-31} kg

h = Planck constant = 6.6×10^{-34} J s

k_B = Boltzmann constant = 1.38×10^{-23} JK^{-1}] **[IIT-Main-2021]**

(A) 6.26 nm (B) 2.26 nm (C) 8.46 nm (D) 3.25 nm

32. Consider the following statements

A. Atoms of each element emit characteristics spectrum.

B. According to Bohr's Postulate, an electron in a hydrogen atom, revolves in a certain stationary orbit.

C. The density of nuclear matter depends on the size of the nucleus.

D. A free neutron is stable but a free proton decay is possible.

E. Radioactivity is an indication of the instability of nuclei.

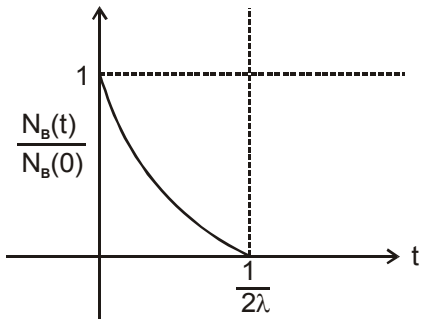
Choose the correct answer from the options given below : **[IIT-Main-2021]**

(A) A, B, C, D and E (B) A, C and E only (C) B and D only (D) A, B and E only

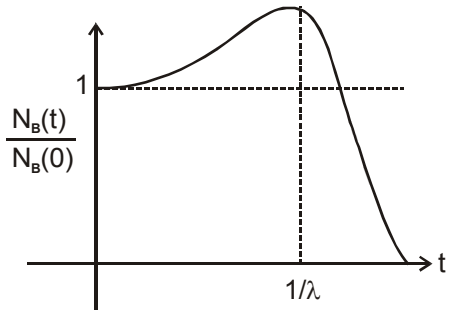
33. At time t = 0, a material is composed of two radioactive atoms A and B, where $N_A(0) = 2N_B(0)$. The decay constant of both kind of radioactive atoms is λ. However, A disintegrates to B and B disintegrates to C. Which of the following figures represents the evolution of $N_B(t)/N_B(0)$ with respect to time t ?

[$N_A(0)$ = No. of A atoms at t = 0, $N_B(0)$ = No. of B atoms at t = 0] **[IIT-Main-2021]**

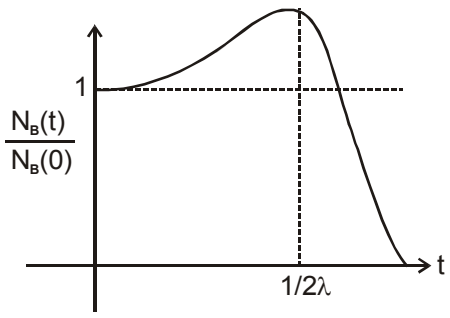
(A)



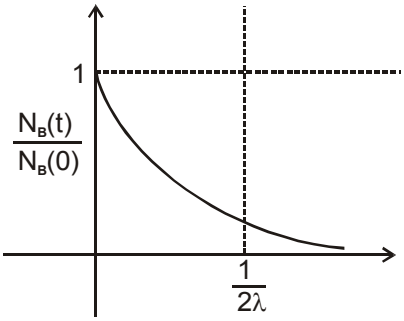
(B)



(C)



(D)



34. A radioactive material decays by simultaneous emissions of two particles with half lives of 1400 years and 700 years respectively. What will be the time after which one third of the material remains? (Take ln 3 = 1.1)

[IIT-Main-2021]

(A) 340 years (B) 700 years (C) 1110 years (D) 740 years

35. If *'f'* denotes the ratio of the number of nuclei decayed (N_d) to the number of nuclei at t = 0 (N_0) then for a collection of radioactive nuclei, the rate of charnge of 'f' with respect to time is given as :
[λ is the radioactive decay constant] **[IIT-Main-2021]**

(A) $\lambda\left(1-e^{-\lambda t}\right)$ (B) $\lambda e^{-\lambda t}$ (C) $-\lambda e^{-\lambda t}$ (D) $-\lambda(1-e^{-\lambda t})$

36. Some nuclei of a radioactive material are undergoing radioactive decay. The time gap between the instances when a quarter of the nuclei have decayed and when half of the nuclei have decayed is given as :
(where λ is the decay constant) **[IIT-Main-2021]**

(A) $\frac{1}{2}\frac{\ln 2}{\lambda}$ (B) $\frac{\ln\frac{3}{2}}{\lambda}$ (C) $\frac{2\ln 2}{\lambda}$ (D) $\frac{\ln 2}{\lambda}$

37. X different wavelengths may be observed in the spectrum from a hydrogen sample if the atoms are exited to states with principal quantum number n = 6 ? The value of X is _____ . **[IIT-Main-2021]**

38. A heavy nucleus Q of half-life 20 minutes undergoes alpha-decay with probability of 60% and beta-decay with probability of 40%. Initially, the number of Q nuclei is 1000. The number of alpha-decays of Q in the first one hour is **[IIT-Advanced-2021]**

(A) 50 (B) 75 (C) 350 (D) 525

39. Which of the following statement(s) is(are) correct about the spectrum of hydrogen atom?
(A) The ratio of the longest wavelength to the shortest wavelength in Balmer series is 9/5
(B) There is an overlap between the wavelength ranges of Balmer and Paschen series
(C) The wavelengths of Lyman series are given by $\left(1+\frac{1}{m^2}\right)\lambda_0$, where λ_0 is the shortest wavelength of Lyman series and m is an integer
(D) The wavelength ranges of Lyman and Balmer series do not overlap **[IIT-Advanced-2021]**

40. A heavy nucleus N, at rest, undergoes fission $N \rightarrow P + Q$, where P and Q are two lighter nuclei. Let $\delta = M_N - M_P - M_Q$, where M_P, M_Q and M_N are the masses of P, Q and N, respectively. E_P and E_Q are the kinetic energies of P and Q, respectively. The speeds of P and Q are v_P and v_Q, respectively. If c is the speed of light, which of the following statement(s) is(are) correct? **[IIT-Advanced-2021]**
(A) $E_P + E_Q = c^2\delta$
(B) $E_P = \left(\frac{M_P}{M_P + M_Q}\right)c^2\delta$
(C) $\frac{v_P}{v_Q} = \frac{M_Q}{M_P}$
(D) The magnitude of momentum for P as well as Q is $c\sqrt{2\mu\delta}$, where $\mu = \frac{M_P M_Q}{(M_P + M_Q)}$

41. An α-particle (mass 4 amu) and a singly charged sulfur ion (mass 32 amu) are initially at rest. They are accelerated through a potential V and then allowed to pass into a region of uniform magnetic field which is normal to the velocities of the particles. Within this region, the α-particle and the sulfur ion move in circular orbits of radii r_α and r_S, respectively. The ratio (r_S/r_α) is _____. **[IIT-Advanced-2021]**

42. In a photoemission experiment, the maximum kinetic energies of photoelectrons from metals P, Q and R are E_P, E_Q and E_R, respectively, and they are related by $E_P = 2E_Q = 2E_R$. In this experiment, the same source of monochromatic light is used for the metal P and Q while a different source of monochromatic light is used for the metal R. The work functions for metals P, Q and R are 4.0 eV, 4.5 eV and 5.5 eV, respectively. The energy of the incident photon used for metal R, in eV, is ..... . **[IIT-Advanced-2021]**

43. The momentum of an electron revolving in n^{th} orbit is given by : (Symbols have their usual meanings) **[IIT-Main-2022]**

(A) $\frac{nh}{2\pi r}$ (B) $\frac{nh}{2r}$ (C) $\frac{nh}{2\pi}$ (D) $\frac{2\pi r}{nh}$

44. Hydrogen atom from excited state comes to the ground by emitting a photon of wavelength λ. The value of principal quantum number 'n' of the excited state will be (R : Rydberg constant) **[IIT-Main-2022]**

(A) $\sqrt{\frac{\lambda R}{\lambda - 1}}$ (B) $\sqrt{\frac{\lambda R}{\lambda R - 1}}$ (C) $\sqrt{\frac{\lambda}{\lambda R - 1}}$ (D) $\sqrt{\frac{\lambda R^2}{\lambda R - 1}}$

45. Find the ratio of energies of photons produced due to transition of an election of hydrogen atom from its (i) second permitted energy level to the first level, and (ii) the highest permitted energy level to the first permitted level. **[IIT-Main-2022]**

(A) 3 : 4 (B) 4 : 3 (C) 1 : 4 (D) 4 : 1

46. An α particle and a proton are accelerated from rest through the same potential difference. The ratio of linear momenta acquired by above two particals will be : **[IIT-Main-2022]**

(A) $\sqrt{2}:1$ (B) $2\sqrt{2}:1$ (C) $4\sqrt{2}:1$ (D) 8 : 1

47. A nucleus of mass M at rest splits into parts having masses $\frac{M'}{3}$ and $\frac{2M'}{3}$ (M' < M). The ratio of de Broglie wavelength of two parts will be **[IIT-Main-2022]**

(A) 1 : 2 (B) 2 : 1 (C) 1 : 1 (D) 2 : 3

48. Read the following statements :

(A) Volume of the nucleus is directly proportional to the mass number.

(B) Volume of the nucleus is independent of mass number.

(C) Density of the nucleus is directly proportional to the mass number.

(D) Density of the nucleus is directly proportional to the cube root of the mass number.

(E) Density of the nucleus is independent of the mass number.

Choose the correct option from the following options. **[IIT-Main-2022]**

(A) (A) and (D) only. (B) (A) and (E) only.

(C) (B) and (E) only. (D) (A) and (C) only

49. The activity of a radioactive material is 6.4×10^{-4} curie. Its half life is 5 days. The activity will become 5×10^{-6} curie after : **[IIT-Main-2022]**

(A) 7 days (B) 15 days (C) 25 days (D) 35 days

50. What is the half-life period of a radioactive material if its activity drops to $1/16^{th}$ of its initial value of 30 years? **[IIT-Main-2022]**

(A) 9.5 years (B) 8.5 years (C) 7.5 years (D) 10.5 years

51. The half life period of a radioactive substance is 60 days. The time taken for $\frac{7}{8}$ th of its original mass to disintegrate will be : **[IIT-Main-2022]**

(A) 120 days (B) 130 days (C) 180 days (D) 20 days

52. Mass numbers of two nuclei are in the ratio of 4 : 3. Their nuclear densities will be in the ratio of **[IIT-Main-2022]**

(A) 4 : 3 (B) $\left(\frac{3}{4}\right)^{1/3}$ (C) 1 : 1 (D) $\left(\frac{4}{3}\right)^{1/3}$

53. The minimum kinetic energy needed by an alpha particle to cause the nuclear reaction ${}^{16}_{7}N + {}^{4}_{2}He \rightarrow {}^{1}_{1}H + {}^{19}_{8}O$ in a laboratory frame is n (in MeV). Assume that ${}^{16}_{7}N$ is at rest in the laboratory frame. The masses of ${}^{16}_{7}N$, ${}^{4}_{2}He$, ${}^{1}_{1}H$ and ${}^{19}_{8}O$ can be taken to be 16.006 u, 4.003 u, 1.008 u and 19.003 u, respectively, where 1 u = 930 MeVc^{-2}. The value of n is _____. **[IIT-Advanced-2022]**

54. The binding energy of nucleons in a nucleus can be affected by the pairwise Coulomb repulsion. Assume that all nucleons are uniformly distributed inside the nucleus. Let the binding energy of a proton be E_b^p and the binding energy of a neutron be E_b^n in the nucleus. **[IIT-Advanced-2022]**

Which of the following statement(s) is(are) correct?

(A) $E_b^p - E_b^n$ is proportional to Z(Z – 1) where Z is the atomic number of the nucleus.

(B) $E_b^p - E_b^n$ is proportional to $A^{-\frac{1}{3}}$ where A is the mass number of the nucleus.

(C) $E_b^p - E_b^n$ is positive.

(D) E_b^p increases if the nucleus undergoes a beta decay emitting a positron.

55. A small particle of mass m moves in such a way that its potential energy U = $\frac{1}{2}$ m $\omega^2 r^2$ where ω is constant and r is the distance of the particle from origin. Assuming Bohr's quantization of momentum and circular orbit, the radius of nth orbit will be proportional to. **[IIT-Main-2023]**

(A) n^2 (B) $\sqrt{n}$ (C) $\frac{1}{n}$ (D) n

56. A 12.5 eV electron beam is used to bombard gaseous hydrogen at room temperature. The number of spectral lines emitted will be : **[IIT-Main-2023]**

(A) 4 (B) 1 (C) 2 (D) 3

57. The energy of He^+ ion in its first excited state is, (The ground state energy for the Hydrogen atom is –13.6 eV) **[IIT-Main-2023]**

(A) –13.6 eV (B) –3.4 eV (C) –54.4 eV (D) –27.2 eV

58. A metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V_0. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential becomes $V_0/4$. The threshold wavelength for this metallic surface will be : **[IIT-Main-2023]**

(A) 3λ (B) 4λ (C) $\frac{3}{2}\lambda$ (D) $\frac{\lambda}{4}$

59. Given below are two statement :

Statement I : Out of microwaves, infrared rays and ultraviolet rays, ultraviolet rays are the most effective for the emission of electrons from a metallic surface.

Statement : II Above the threshold frequency, the maximum kinetic energy of photoelectrons is inversely proportional to the frequency of the incident light.

In the light of above statements, choose the correct answer from the options given below **[IIT-Main-2023]**

(A) Statement I is true but statement II is false

(B) Both statement I and statement II are true

(C) statement I is false but statement II is true

(D) Both Statement I and Statement II are false

60. The difference between threshold wavelengths for two metal surfaces A and B having work function $\phi_A = 9$ eV and $\phi_B = 4.5$ eV in nm is : {Given, hc = 1242 eV nm} **[IIT-Main-2023]**

(A) 540 (B) 276 (C) 264 (D) 138

61. For a nucleus ${}^{A}_{Z}X$ having mass number A and atomic number Z

(A) The surface energy per nucleon $(b_S) = -a_1 A^{2/3}$

(B) The Coulomb contribution to the binding energy $b_c = -a_2 \frac{Z(Z-1)}{A^{4/3}}$

(C) The volume energy $b_v = a_3 A$

(D) Decrease in the binding energy is proportional to surface area.

(E) While estimating the surface energy, it is assumed that each nucleon interacts with 12 nucleons. (a_1, a_2 and a_3 are constants)

Choose the most appropriate answer from the options given below : **[IIT-Main-2023]**

(A) C, D only (B) A, B, C, D only (C) B, C only (D) B, C, E only

62. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : The binding energy per nucleon is practically indepedent of the atomic number for nuclei of mass number in the range 30 to 170.

Reason R : Nuclear force is short ranged.

In the light of the above statements, choose the correct answer from the options given below **[IIT-Main-2023]**

(A) Both A and R true but R is NOT the correct explanation of A

(B) Both A and R are true and R is the correct explanation of A

(C) A is false but R is true

(D) A is true but R is false

63. The half-life of a radioactive nucleus is 5 years. The fraction of the original sample that would decay in 15 years is : **[IIT-Main-2023]**

(A) $\frac{3}{4}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{7}{8}$

64. Two radioactive elements A and B initially have same number of atoms. The half life of A is same as the average life of B. If λ_A and λ_B are decay constants of A and B respectively, then choose the correct relation from the given options : **[IIT-Main-2023]**

(A) $\lambda_A \ell n2 = \lambda_B$ (B) $\lambda_A = \lambda_B$ (C) $\lambda_A = 2\lambda_B$ (D) $\lambda_A = \lambda_B \ell n2$

65. A Hydrogen-like atom has atomic number Z. Photons emitted in the electronic transitions from level n = 4 to level n = 3 in these atoms are used to perform photoelectric effect experiment on a target metal. The maximum kinetic energy of the photoelectrons generated is 1. 95 eV. If the photoelectric threshold wavelength for the target metal is 310 nm, the value of Z is _____. **[IIT-Advanced-2023]**

[Given : hc = 1240 eV-nm and Rhc = 13.6 eV, where R is the Rydberg constant, h is the Planck's constant and c is the speed of light in vacuum]

66. List-I shows different radioactive decay processes and List-II provides possible emitted particles. Match each entry in List-I with an appropriate entry from List-II, and choose the correct option. **[IIT-Advanced-2023]**

List-I	List-II
(P) $^{238}_{92}U \rightarrow {}^{234}_{91}Pa$	(1) one α particle and one β^+ particle
(Q) $^{214}_{82}Pb \rightarrow {}^{210}_{82}Pb$	(2) three β^- particles and one α particle
(R) $^{210}_{81}T\ell \rightarrow {}^{206}_{82}Pb$	(3) two β^- particles and one α particle
(S) $^{228}_{91}Pa \rightarrow {}^{224}_{88}Ra$	(4) one α particle and one β^- particle
	(5) one α particle and two β^+ particles

(A) P $\rightarrow$ 4, Q $\rightarrow$ 3, R $\rightarrow$ 2, S $\rightarrow$ 1 (B) P $\rightarrow$ 4, Q $\rightarrow$ 1, R $\rightarrow$ 2, S $\rightarrow$ 5

(C) P $\rightarrow$ 5, Q $\rightarrow$ 3, R $\rightarrow$ 1, S $\rightarrow$ 4 (D) P $\rightarrow$ 5, Q $\rightarrow$ 1, R $\rightarrow$ 3, S $\rightarrow$ 2

67. Match the temperature of a black body given in List-I with an appropriate statement in List-II, and choose the correct option. **[IIT-Advanced-2023]**

[Given : Wien's constant as 2.9×10^{-3} m-K and $\frac{hc}{e} = 1.24 \times 10^{-6}$ V-m]

List-I	List-II
(P) 2000 K	(1) The radiation at peak wavelength can lead to emission of photoelectrons from a metal of work function 4 eV.
(Q) 3000 K	(2) The radiation at peak wavelength is visible to human eye.
(R) 5000 K	(3) The radiation at peak emission wavelength will result in the widest central maximum of a single slit diffraction.
(S) 10000 K	(4) The power emitted per unit area is 1/16 of that emitted by a blackbody at temperature 6000 K.
	(5) The radiation at peak emission wavelength can be used to image human bones.

(A) P $\rightarrow$ 3, Q $\rightarrow$ 5, R $\rightarrow$ 2, S $\rightarrow$ 3 (B) P $\rightarrow$ 3, Q $\rightarrow$ 2, R $\rightarrow$ 4, S $\rightarrow$ 1

(C) P $\rightarrow$ 3, Q $\rightarrow$ 4, R $\rightarrow$ 2, S $\rightarrow$ 1 (D) P $\rightarrow$ 1, Q $\rightarrow$ 2, R $\rightarrow$ 5, S $\rightarrow$ 3

68. In a radioactive decay process, the activity is defined as $A = -\frac{dN}{dt}$, where N(t) is the number of radioactive nuclei at time t. Two radioactive sources, S_1 and S_2 have same activity at time t = 0. At a later time, the activities of S_1 and S_2 are S_2 are A_1 and A_2 respectively. When S_1 and S_2 have just completed their 3rd and 7th half-lives, respectively, the ratio A_1/A_2 is _____. **[IIT-Advanced-2023]**

69. A hydrogen atom in ground state is given an energy of 10.2 eV. How many spectral lines will be emitted due to transition of electrosn ? **[IIT-Main-2024]**

(A) 10 (B) 6 (C) 3 (D) 1

70. The longest wavelength associated with Paschen series is : (Given $R_H = 1.097 \times 10^7$ SI unit) **[IIT-Main-2024]**

(A) 3.646×10^{-6} m (B) 1.094×10^{-6} m (C) 2.973×10^{-6} (D) 1.876×10^{-6} m

71. The ratio of the shortest wavelength of Balmer series to the shortest wavelength of Lyman series for hydrogen atom is : **[IIT-Main-2024]**

(A) 1 : 4 (B) 4 : 1 (C) 2 : 1 (D) 1 : 2

72. According to Bohr's theory, the moment of momentum of an electron revolving in 4th orbit of hydrogen atom is : **[IIT-Main-2024]**

(A) $2\frac{h}{\pi}$ (B) $\frac{h}{\pi}$ (C) $\frac{h}{2\pi}$ (D) $8\frac{h}{\pi}$

73. UV light of 4.13 eV is incidenty on a photosensitive metal surface having work function 3.13 eV. The maximum kinetic energy of ejected photoelectrons will be : **[IIT-Main-2024]**

(A) 4.13 eV (B) 7.26 eV (C) 3.13 eV (D) 1 eV

74. A proton, an electron and an alpha particle have the same energies. Their de-Broglie wavelengths will be compared as : **[IIT-Main-2024]**

(A) $\lambda_\alpha < \lambda_p < \lambda_e$ (B) $\lambda_p > \lambda_e > \lambda_\alpha$ (C) $\lambda_p < \lambda_e < \lambda_\alpha$ (D) $\lambda_e > \lambda_\alpha > \lambda_p$

75. A proton and an electron have the same de Broglie wavelength. If K_p and K_e be the kinetic energies of proton and electron respectively, then choose the correct relation : **[IIT-Main-2024]**

(A) $K_p < K_e$ (B) $K_p = K_e^2$ (C) $K_p = K_e$ (D) $K_p > K_e$

76. The energy released in the fusion of 2 kg of hydrogen deep in the sun in E_H and the energy released in the fission of 2 kg of ^{235}U is E_U. The ratio $\frac{E_H}{E_U}$ is approximately:

(consider the fusion reaction as $4\,H + 2e^- \rightarrow {}_2^4He + 2\nu + 5\gamma + 26.7$ MeV, energy released in the fission reaction of ^{235}U is 200 MeV per fission nucleus and $N_A = 6.023 \times 10^{23}$) **[IIT-Main-2024]**

(A) 9.13 (B) 15.04 (C) 7.62 (D) 25.6

77. A nucleus at rest disintegrates into two smaller nuclei with their mases in the ratio of 2:1. After disintegration they will move : **[IIT-Main-2024]**

(A) in opposite directions with speed in the ratio of 1:2 respectively.
(B) in opposite directions with the same speed.
(C) in the same direction with same speed.
(D) in opposite directions with speed in the ratio of 2:1 respectively.

78. The energy equivalent of 1 g of substance is : **[IIT-Main-2024]**

(A) 5.6×10^{26} MeV (B) 5.6 eV (C) 11.2×10^{24} MeV (D) 5.6×10^{12} MeV

79. A metal target with atomic number Z = 46 is bombarded with a high energy electron beam. The emission of X-rays from the target is analyzed. The ratio r of the wavelengths of the K_α-line and the cut-off is found to be r = 2. If the same electron beam bombards another metal target with Z = 41, the value of r will be **[IIT-Advanced-2024]**

(A) 2.53 (B) 1.27 (C) 2.24 (D) 1.58

80. A light source of wavelength λ illuminates a metal surface and electrons are ejected with maximum kinetic energy of 2 eV. If the same surface is illuminated by a light source of wavelength $\frac{\lambda}{2}$, then the maximum kinetic energy of ejected electrons will be (The work function of metal is 1 eV) **[IIT-Main-2025]**

(A) 2 eV (B) 6 eV (C) 5 eV (D) 3 eV

81. In photoelectric effect an em-wave is incident on a metal surface and electrons are ejected from the surface. If the work function of the metal is 2.14 eV and stopping potential is 2V, what is the wavelength of the em-wave? (Given hc = 1242 eVnm where h is the Planck's constant and c is the speed of light in vaccum.) **[IIT-Main-2025]**

(A) 400 nm (B) 600 nm (C) 200 nm (D) 300 nm

82. A sub-atomic particle of mass 10^{-30} kg is moving with a velocity 2.21×10^6 m/s. Under the matter wave consideration, the particle will behave closely like _____. ($h = 6.63 \times 10^{-34}$ J.s) **[IIT-Main-2025]**

(A) Infra-red radiation (B) X-rays (C) Gamma rays (D) Visible radiation

83. The work functions of cesium (Cs) and lithium (Li) metals are 1.9 eV and 2.5 eV, respectively. If we incident a light of wavelength 550 nm on these two metal surface, then photo-electric effect is possible for the case of **[IIT-Main-2025]**

(A) Li only (B) Cs only (C) Neither Cs nor Li (D) Both Cs and Li

84. In photoelectric effect, the stopping potential (V_0) v/s frequency (ν) curve is plotted.
(h is the Planck's constant and ϕ_0 is work function of metal)

(I) V_0 v/s ν is linear

(III) The slope of V_0 v/s ν curve = $\frac{\phi_0}{h}$

(III) h constant is related to the slope of V_0 v/s ν line

(IV) The value of electric charge of electron is not required to determine h using the V_0 v/s ν curve.

(V) The work function can be estimated without knowing the value of h.

Choose the **correct** answer from the options given below : **[IIT-Main-2025]**

(A) (I),(II) and (III) only (B) (III) and (IV) only (C) (I),(III) and (V) only (D) (IV) and (V) only

85. The energy E and momentum p of a moving body of mass m are related by some equation. Given that c represents the speed of light, identify the correct equation. **[IIT-Main-2025]**

(A) $E^2 = pc^2 + m^2c^4$ (B) $E^2 = pc^2 + m^2c^2$ (C) $E^2 = p^2c^2 + m^2c^2$ (D) $E^2 = p^2c^2 + m^2c^4$

86. In an experiment with photoelectric effect, the stopping potential. **[IIT-Main-2025]**

(A) increases with increase in the wavelength of the incident light

(B) increases with increase in the intensity of the incident light

(C) is $\left(\frac{1}{e}\right)$ times the maximum kinetic energy of the emitted photoelectrons

(D) decreases with increase in the intensity of the incident light

87. Which of the following phenomena can not be explained by wave theory of light? **[IIT-Main-2025]**

(A) Reflection of light (B) Diffraction of light

(C) Refraction of light (D) Compton effect

88. Given below are two statements : one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.
Assertion (A) : Electromagnetic waves carry energy but not momentum.
Reason (R) : Mass of a photon is zero.
In the light of the above statements, choose the **most appropriate answer** from the options given below :
[IIT-Main-2025]

(A) (A) is true but (R) is false.
(B) (A) is false but (R) is true.
(C) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(D) Both (A) and (R) are true and (R) is the correct explanation of (A)

89. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Emission of electrons in photoelectric effect can be suppressed by applying a sufficiently negative electron potential to the photoemissive substance.
Reason (R) : A negative electric potential, which stops the emission of electrons from the surface of a photoemissive substance, varies linearly with frequency of incident radiation.
In the light of the above statements, choose the **most appropriate answer** from the options given below :
[IIT-Main-2025]

(A) (A) is false but (R) is true.
(B) (A) is true but (R) is false.
(C) Both (A) and (R) are true and (R) is the correct explanation of (A).
(D) Both (A) and (R) are true but (R) is not the correct explanation of (A)

90. A monochromatic light is incident on a metallic plate having work function ϕ. An electron, emitted normally to the plate from a point A with maximum kinetic energy, enters a constant magnetic field, perpendicular to the initial velocity of electron. The electron passes through a curve and hits back the plate at a point B. The distance between A and B is :
(Given : The magnitude of charge of an electron is e and mass is m, h is Planck's constant and c is velocity of light. Take the magnetic field exists throughout the path of electron) **[IIT-Main-2025]**

(A) $\sqrt{2m\left(\frac{hc}{\lambda}-\phi\right)}/eB$ (B) $\sqrt{m\left(\frac{hc}{\lambda}-\phi\right)}/eB$ (C) $\sqrt{8m\left(\frac{hc}{\lambda}-\phi\right)}/eB$ (D) $2\sqrt{m\left(\frac{hc}{\lambda}-\phi\right)}/eB$

91. Match the **LIST-I** with **LIST-II**

LIST-I	**LIST-II**
(a) Boltzmann constant	I. ML^2T^{-1}
(b) Coefficient of viscosity	II. $MLT^{-3}K^{-1}$
(c) Planck's constant	III. $ML^2T^{-2}K^{-1}$
(d) Thermal conductivity	IV. $ML^{-1}T^{-1}$

Choose the correct answer from the options given below : **[IIT-Main-2025]**
(A) a-III, b-IV, c-I, d-II (B) a-II, b-III, c-IV, d-I
(C) a-III, b-II, c-I, d-IV (D) a-III, b-IV, c-II, d-I

92. The radiation pressure exerted by a 450 W light source on a perfectly reflecting surface placed at 2m away from it, is : **[IIT-Main-2025]**
(A) 1.5×10^{-8} Pascals (B) 0 (C) 6×10^{-8} Pascals (D) 3×10^{-8} Pascals

93. The work function of a metal is 3 eV. The color of the visible light that is required to cause emission of photoelectrons is : **[IIT-Main-2025]**
(A) Green (B) Blue (C) Red (D) Yellow

94. A small mirror of mass m is suspended by a massless thread of length ℓ. Then the small angle through which the thread will be deflected when a short pulse of laser of energy E falls normal on the mirror (c = speed of light in vacuum and g = acceleration due to gravity) **[IIT-Main-2025]**

(A) $\theta = \frac{3E}{4mc\sqrt{g\ell}}$ (B) $\theta = \frac{E}{mc\sqrt{g\ell}}$ (C) $\theta = \frac{E}{2mc\sqrt{g\ell}}$ (D) $\theta = \frac{2E}{mc\sqrt{g\ell}}$

95. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**
Assertion A : In photoelectric effect, on increasing the intensity of incident light the stopping potential increases.
Reason R : Increase in intensity of light increases the rate of photoelectrons emitted, provided the frequency of incident light is greater than threshold frequency.
In the light of the above statements, choose the correct answer from the options given below **[IIT-Main-2025]**

(A) Both A and R are true but R is NOT the correct explanation of A
(B) A is false but R is true
(C) A is true but R is false
(D) Both A and R are true and R is the correct explanation of A

96. A photo-emissive substance is illuminated with a radiation of wavelength λ_i so that it releases electrons with de-Broglie wavelength λ_e. The longest wavelength of radiation that can emit photoelectron is λ_0. Expression for de-Broglie wavelength is given by :
(m : mass of the electron, h : Planck's constant and c : speed of light) **[IIT-Main-2025]**

(A) $\lambda_e = \sqrt{\frac{h}{2mc\left(\frac{1}{\lambda_i} - \frac{1}{\lambda_0}\right)}}$ (B) $\lambda_e = \sqrt{\frac{h\lambda_0}{2mc}}$

(C) $\lambda_e = \frac{h}{\sqrt{2mc\left(\frac{1}{\lambda_i} - \frac{1}{\lambda_0}\right)}}$ (D) $\lambda_e = \sqrt{\frac{h\lambda_i}{2mc}}$

97. Choose the correct nuclear process from the below options
[p : proton, n : neutron, e^- : electron, e^- : positron, v : neutrino, $\bar{v}$: antineutrino] **[IIT-Main-2025]**

(A) $n \to p + e^- + \bar{v}$ (B) $n \to p + e^- + v$ (C) $n \to p + e^+ + \bar{v}$ (D) $n \to p + e^+ + v$

98. A radioactive nucleus n_2 has 3 times the decay constant as compared to the decay constant of another radioactive nucleus n_1. If initial number of both nuclei are the same, what is the ratio of number of nuclei of n_2 to the number of nuclei of n_1, after one half-life of n_1 ? **[IIT-Main-2025]**
(A) 1/4 (B) 1/8 (C) 4 (D) 8

99. Given below are two statements. One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : The binding energy per nucleon is found to be practically independent of the atomic number A, for nuclei with mass numbers between 30 and 170.
Reason (R) : Nuclear force is long range. **[IIT-Main-2025]**
In the light of the above statements, choose the **correct** answer from the options given below :
(A) (A) is false but (R) is true
(B) (A) is true but (R) is false
(C) Both (A) and (R) are true and (R) is the correct explanation of (A)
(D) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

100. Energy released when two deuterons ($_1H^2$) fuse to form a helium nucleus ($_2He^4$) is :
(Given : Binding energy per nucleon of $_1H^2$ = 1.1 MeV and binding energy per nucleon of $_2He^4$ = 7.0 MeV)
[IIT-Main-2025]

(A) 8.1 MeV (B) 5.9 MeV (C) 23.6 MeV (D) 26.8 MeV

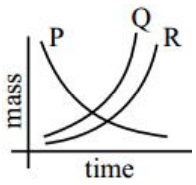
101. Match the **List-I** with **List-II**

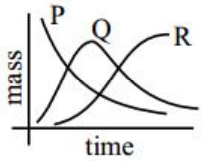
List-I	**List-II**
(a) $^1_0n + {}^{235}_{92}U \rightarrow {}^{140}_{54}Xe + {}^{94}_{38}Sr + 2{}^1_0n$	(I) Chemical reaction
(b) $2H_2 + O_2 \rightarrow 2H_2O$	(II) Fusion with +ve Q value
(c) $^2_1H + {}^2_1H \rightarrow {}^3_2He + {}^1_0n$	(III) Fission
(d) $^1_1H + {}^3_1H \rightarrow {}^2_1H + {}^2_1H$	(IV) Fusion with –ve Q value

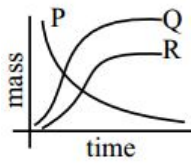
Choose the **correct** answer from the options given below : **[IIT-Main-2025]**

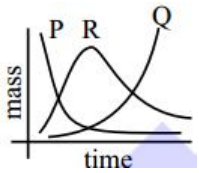
(A) a-II, b-I, c-III, d-IV (B) a-III, b-I, c-II, d-IV
(C) a-II, b-I, c-IV, d-III (D) a-III, b-I, c-IV, d-II

102. A radioactive material P first decays into Q and then Q decays to non-radioactive material R. Which of the following figure represents time dependent mass of P, Q and R? **[IIT-Main-2025]**

(A) 

(B) 

(C) 

(D) 

103. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as Reason **(R)**.

Assertion (A) : The density of the copper $(^{64}_{29}Cu)$ nucleus is greater than that of the carbon $(^{12}_{6}C)$ nucleus.

Reason (R) : The nucleus of mass number A has a radius proportional to $A^{1/3}$.
In the light of the above statements, choose the most appropriate answer from the options given below :
[IIT-Main-2025]

(A) **(A)** is correct but **(R)** is not correct
(B) **(A)** is not correct but **(R)** is correct
(C) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
(D) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

104. For a nucleus of mass number A and radius R, the mass density of nucleus can be represented as
[IIT-Main-2025]

(A) A^3 (B) $A^{1/3}$ (C) $A^{2/3}$ (D) Independent of A

105. The ratio of the power of a light source S_1 to that the light source S_2 is 2. S_1 is emitting 2×10^{15} photons per second at 600 nm. If the wavelength of the source S_2 is 300 nm, then the number of photons per second emitted by S_2 is_____$\times 10^{14}$. **[IIT-Main-2025]**

106. **List-I** shows various functional dependencies of energy (E) on the atomic number (Z). Energies associated with certain phenomena are given in **List-II**.
Choose the option that describes the correct match between the entries in **List-I** to those in **List-II**.

[IIT-Advanced-2025]

List-I		**List-II**
(P) $E \propto Z^2$	(1)	Energy of characteristic x-rays
(Q) $E \propto (Z-1)^2$	(2)	Electrostatic part of the nuclear binding energy for stable nuclei with mass numbers in the range 30 to 170
(R) $E \propto Z(Z-1)$	(3)	Energy of continuous x-rays
(S) E is practically independent of Z	(4)	Average nuclear binding energy per nucleon for stable nuclei with mass number in the range 30 to 170
	(5)	Energy of radiation due to electronic transitions from hydrogen-like atoms

(A) P $\rightarrow$ 4, Q $\rightarrow$ 3, R $\rightarrow$ 1, S $\rightarrow$ 2
(B) P $\rightarrow$ 5, Q $\rightarrow$ 2, R $\rightarrow$ 1, S $\rightarrow$ 4
(C) P $\rightarrow$ 5, Q $\rightarrow$ 1, R $\rightarrow$ 2, S $\rightarrow$ 4
(D) P $\rightarrow$ 3, Q $\rightarrow$ 2, R $\rightarrow$ 1, S $\rightarrow$ 5

107. Consider an electron in the n = 3 orbit of a hydrogen-like atom with atomic number Z. At absolute temperature T, a neutron having thermal energy k_BT has the same de Broglie wavelength as that of this electron. If this temperature is given by $T = \frac{Z^2h^2}{\alpha\pi^2a_0^2m_Nk_B}$, (where h is the Planck's constant, k_B is the Boltzmann constant, m_N is the mass of the neutron and a_0 is the first Bohr radius of hydrogen atom) then the value of α is ___.

[IIT-Advanced-2025]

108. A hydrogen atom, initially at rest in its ground state, absorbs a photon of frequency ν_1 and ejects the electron with a kinetic energy of 10 eV. The electron then combines with a positron at rest to form a positronium atom in its ground state and simultaneously emits a photon of frequency ν_2. The center of mass of the resulting positronium atom moves with a kinetic energy of 5 eV. It is given that positron has the same mass as that of electron and the positronium atom can be considered as a Bohr atom, in which the electron and the positron orbit around their center of mass. Considering no other energy loss during the whole process, the difference between the two photon energies (in eV) is ____. **[IIT-Advanced-2025]**

ANSWER KEY

EXERCISE-I

ONLY ONE OPTION IS CORRECT

1.	D	**2.**	A	**3.**	B	**4.**	C	**5.**	D	**6.**	C	**7.**	A
8.	C	**9.**	C	**10.**	D	**11.**	C	**12.**	C	**13.**	A	**14.**	C
15.	B	**16.**	B	**17.**	B	**18.**	A	**19.**	B	**20.**	B	**21.**	C
22.	C	**23.**	B	**24.**	B	**25.**	C	**26.**	A	**27.**	D	**28.**	A
29.	C	**30.**	D	**31.**	C	**32.**	D	**33.**	C	**34.**	B	**35.**	A
36.	A	**37.**	A	**38.**	A	**39.**	B	**40.**	D	**41.**	C	**42.**	C
43.	A	**44.**	C	**45.**	D	**46.**	C	**47.**	C	**48.**	C	**49.**	B
50.	A	**51.**	B	**52.**	D	**53.**	B	**54.**	C	**55.**	B	**56.**	D
57.	A	**58.**	B	**59.**	C	**60.**	B	**61.**	B	**62.**	D	**63.**	C
64.	A	**65.**	C										

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

66.	A, C	**67.**	D	**68.**	B	**69.**	B	**70.**	B	**71.**	A	**72.**	A CD
73.	B	**74.**	A, B	**75.**	C	**76.**	A	**77.**	A, C	**78.**	A, B, D	**79.**	A
80.	B	**81.**	A, D	**82.**	A, B	**83.**	D	**84.**	C	**85.**	A, C	**86.**	C, D
87.	B	**88.**	C	**89.**	D								

90. (i) – A, (ii) – E, (iii) – F, (iv) – C,

EXERCISE-II

1. (a) 2.25eV, (b) 4.2eV, (c) 2.0 eV, 0.5 eV

2. (a) 0.6 volt, (b) 2.0 mA

3. when the potential is steady, photo electric emission just stop when $h\upsilon = (3 + 1)eV = 4.0$ eV

4. 5.76×10^{-11} A

5. 15/8 V

6. $\frac{5}{16}, \frac{10^{20}}{80\pi}$

7. 2 eV, 6.53×10^{-34} J-s

8. 885

9. $-h/eEt^2$

10. 487.06 nm

11. 4.26 m/s, 13.2 eV

12. 7 : 36

13. 22.8 nm

14. $\frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}$

15. 18/(5R)

16. 1.257×10^{-23} Am^2

17. 5

18. (i) 5, 16.5 eV, 36.4 A, 340 eV, – 680 eV, $\frac{h}{2\pi}$ 1.06×10^{-11} m

19. z = 3, n = 7

20. 54.4 eV

21. n = 3, 3 : 1

22. 23.6 MeV

23. ($T_{1/2}$ = 10.8 sec)

24. $t = \left(\frac{\ln 5}{\ln 2}\right)\tau$

25. $8/3 \times 10^{18}$sec

26. 2.48×10^{-12} m

27. (i) $\frac{he}{4\pi m}$ (ii) $\frac{ehB}{8\pi m}$

28. 4.8×10^{16}, 4.0 cm

29. 1.1×10^{12}

30. $\lambda_{deutron} = \lambda_{neutron} = 8.6$ pm **31.** $\lambda = \dfrac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2+\lambda_2^2}}$

32. 3.1×10^6 m/s **33.** (i) 2 ; (ii) 23.04×10^{-19} J ; (iii) $4 \to 1$, $4 \to 3$

34. 11.24 eV **35.** 6.8 eV, 5×10^{15} Hz **36.** 489.6 eV, 25.28 Å

37. (i) 1.875×10^4 V, (ii) 2.7×10^{-15} J, (iii) 0.737 Å, (iv) 2.67×10^{-15} J **38.** 6.04×10^9 yrs

39. 4.87 MeV **40.** 3.3×10^{-6} g

41. 5196 yrs **42.** $\Delta T = \dfrac{0.2E_0\left[\alpha t - \dfrac{\alpha}{\lambda}(1-e^{-\lambda t})\right]}{mS}$

43. (a) $10^5\ s^{-1}$; (b) 286.18 ; (d) 111 s **44.** KE $\cong$ 151 eV, d_{least} = 0.5 Å **45.** $\cong$ 33.298 μW

EXERCISE-III

1.	A	**2.**	B	**3.**	A, C	**4.**	A						
5.	A – R, T ; B – P, S ; C – Q, R, T ; D – Q, R												
6.	2	**7.**	3	**8.**	2	**9.**	2	**10.**	B	**11.**	A, B, D	**12.**	C
13.	A	**14.**	6	**15.**	9	**16.**	A	**17.**	5	**18.**	5	**19.**	A,C
20.	24.00	**21.**	3.00	**22.**	C	**23.**	AC	**24.**	135.00 keV (130 – 140)			**25.**	1.00
26.	B, C	**27.**	A, C	**28.**	D	**29.**	A	**30.**	B	**31.**	A	**32.**	D
33.	C	**34.**	D	**35.**	B	**36.**	B	**37.**	15.00	**38.**	D	**39.**	A, D
40.	A,C,D	**41.**	4.00	**42.**	6.00	**43.**	A	**44.**	B	**45.**	A	**46.**	B
47.	C	**48.**	B	**49.**	D	**50.**	C	**51.**	C	**52.**	C	**53.**	2.33
54.	A,B,D	**55.**	B	**56.**	B	**57.**	A	**58.**	A	**59.**	A	**60.**	D
61.	A	**62.**	B	**63.**	D	**64.**	D	**65.**	3.00	**66.**	A	**67.**	C
68.	16.00	**69.**	D	**70.**	D	**71.**	B	**72.**	A	**73.**	D	**74.**	A
75.	A	**76.**	C	**77.**	A	**78.**	A	**79.**	A	**80.**	C	**81.**	D
82.	B	**83.**	B	**84.**	C	**85.**	D	**86.**	C	**87.**	D	**88.**	B
89.	D	**90.**	C	**91.**	A	**92.**	C	**93.**	B	**94.**	D	**95.**	B
96.	A	**97.**	A	**98.**	A	**99.**	B	**100.**	C	**101.**	B	**102.**	B
103.	B	**104.**	D	**105.**	5.00	**106.**	C	**107.**	72.00	**108.**	11.80		